

TLS-SE 3.03 TESTS

Version 1.0

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1 Test Platform

These tests have been performed with the javacard J3R180.

1.1 Downloading TLS-SE.cap

The file winscard.dll is a dynamic SHIM library that enables the use of GPSHELL software with RACS server.

There are two ways for downloading TLS-SE.cap application in Javacard

- To use GPSHELL without RACS server, rename the file winscard.dll) to _winscard.dll
- To use GPSHELL with RACS server, start run_iose.bat

GP3_LIST.bat lists the javacard embedded applications

GP3_Install_tlse.bat loads tls-se.cap in the javacard

2 Tests TLS

These tests require an Internet of Secure Element (IoSE) server using IP loopback address and port 8888 for TLS front server.

To start IoSE server (winscard.dll is required)

- run_iose.bat (console version)
- run_iosew.bat (windows version)

2.1 test_openssl_trait

```
@echo off
set IP=127.0.0.1
set PORT=8888
set SN=key1.com
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set ID=Client_identity
set MYCMD=.%00
REM
echo shells://%SN%:%PSK%:%IP%:%PORT%/?query=%MYCMD%
echo %MYCMD%|openssl s_client -quiet -connect %IP%:%PORT% -servername %SN% -psk_identity %ID%
-psk %PSK% -tls1_3 -groups P-256 -cipher DHE -ciphersuites TLS_AES_128_CCM_SHA256 -no_ticket
2>nul
PAUSE

shells://key1.com:0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20@127.0.0.1:8
888/?query=.%00
Ethertrust keystore 3.03
```

2.2 test_openssl

```
@echo off
set IP=127.0.0.1
set PORT=8888
set SN=key1.com
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set ID=Client_identity
REM
```



```
Device key1.com is authenticated...
ClientSession with key1.com opened in 1860 ms

TLS1.3 Disconnected from key1.com
```

2.4 test_echo

```
@echo off
set PORT=8888
set SEN=key1.com
set IP=127.0.0.1
set nb=001
set
V256=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
REM echo 256 characters
tlsse -c -H @%nb%?01%V256% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
REM echo 512 characters
tlsse -c -H @%nb%?01%V256%%V256% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-
SHA256 -H psk%PSK% -H identity%id%
REM echo 128 bytes hexa coded
tlsse -c -H @%nb%?FF%V256% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 1062 ms
ClientSession with key1.com opened in 1062 ms

Send:
?011234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890a
bcedf1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv:
1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcd
ef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890ab
cdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
in 750 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 980 ms
ClientSession with key1.com opened in 980 ms

Send:
?011234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890a
bcedf1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345
67890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123
4567890abcdef1234567890abcdef1234567890abcdef
Recv:
1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcd
ef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890ab
cdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456
7890abcdef1234567890abcdef1234567890abcdef
in 1187 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 968 ms
ClientSession with key1.com opened in 968 ms

Send:
?FF1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890a
```


ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 926 ms
 ClientSession with key1.com opened in 926 ms

Send:
 ?AA0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F200102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
 Recv: 0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
 in 590 ms
 Send: ?A00102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
 Recv: 0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
 in 250 ms
 Send: ?A1
 Recv: 0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
 in 125 ms
 Send: ?05Client_identity
 Recv: Client_identity
 in 94 ms
 Send: ?06guest
 Recv: guest
 in 78 ms
 Send: ?A80507
 Recv: OK
 in 62 ms
 Send: ?A501
 Recv: OK
 in 78 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 844 ms
 ClientSession with key1.com opened in 844 ms

Send: ?00
 Recv: Ethertrust keystore 3.03
 in 297 ms
 Send: ?FFabcd
 Recv: ABCD
 in 78 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 843 ms
 ClientSession with key1.com opened in 843 ms

Send: ?A80503
 Recv: OK
 in 282 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 844 ms
 ClientSession with key1.com opened in 844 ms

Send: ?FFabcd
 Recv: ERROR
 in 281 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 844 ms
 ClientSession with key1.com opened in 844 ms

Send: ?A500
 Recv: OK
 in 297 ms
 Send: ?A80507
 Recv: OK
 in 62 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888

```
Using Software Identity Module
TLS1.3 Connection Failed with with key1.com
```

2.6 test_console

```
@echo off
set IP=127.0.0.1
set PORT=8888
set SEN=key1.com
set IDENTITY=Client_identity
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set
CAPub=046099836D971593AAA2C1C32B6DB9EF9521041795E21CF1E7511DF3BD358F97DF358B33A875E359CBE23616
3D6DBAEDFEC6C9393522C7EBC25A7CC85E1F0A7D67
REM
tlssse -c -H console -h %IP% -p %PORT% -S %SEN% -H identity%IDENTITY% -H psk%PSK%
PAUSE

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 1057 ms
ClientSession with key1.com opened in 1057 ms
Send (! or : to exit)
?00
Recv:
Ethertrust keystore 3.03
in 78 ms
Send (! or : to exit)
?02
Recv:
TLS1.3 Disconnected from key1.com
```

2.7 test_console_guest

```
@echo off
set IP=127.0.0.1
set PORT=8888
set SEN=key1.com
set IDENTITY=Client_identity
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set
CAPub=046099836D971593AAA2C1C32B6DB9EF9521041795E21CF1E7511DF3BD358F97DF358B33A875E359CBE23616
3D6DBAEDFEC6C9393522C7EBC25A7CC85E1F0A7D67
REM
tlssse -c -H #?A501 -h %IP% -p %PORT% -S %SEN% -H identity%IDENTITY% -H psk%PSK%
set IDENTITY=guest
tlssse -c -H console -h %IP% -p %PORT% -S %SEN% -H identity%IDENTITY% -H psk%PSK%
set IDENTITY=Client_identity
tlssse -c -H #?A500 -h %IP% -p %PORT% -S %SEN% -H identity%IDENTITY% -H psk%PSK%
PAUSE

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 920 ms
ClientSession with key1.com opened in 920 ms

Send: ?A501
Recv: OK
in 281 ms
TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 844 ms
ClientSession with key1.com opened in 844 ms
Send (! or : to exit)
?00
Recv:
Ethertrust keystore 3.03
in 78 ms
Send (! or : to exit)
c00
```

```

Recv:
ERROR
in 78 ms
Send (! or : to exit)
?02
Recv:
TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 859 ms
ClientSession with key1.com opened in 859 ms

Send: ?A500
Recv: OK
in 281 ms
TLS1.3 Disconnected from key1.com

```

2.8 test_set_ecc

```

@echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
set
pub0=0495C60DCFF9636B7603BD4C52B4048C260F9B87E3975F7599A072FCBD25B417ADDEACBD972DFD55771E459CB
50C10079C0AEC0136F822871BC860D771DC554F6C
set priv0=484B1E61536050B59505943FFED7C8A6A5639B2FF7B42600F889B470E67BF89F
set
pub1=0411E4E60C0A2461F3A94FC2F7ACBB6C527855BD08C90B91320AE796178628531033B6C4FCC6C04DD8B278D97
238B6C34501FC120D461A16EBB35756D3CFDD43D2
set priv1=435B6E516AB693CA54267D5BB50956D9C2F54F90449F16D0034A3B2D64EEEF10
REM
set V32=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
REM
tlsse -c -H #c00 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
tlsse -c -H #P00%pub0% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
tlsse -c -H #R00%priv0% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
tlsse -c -H #s00%V32% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
tlsse -c -H #C00 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
tlsse -c -H #P00%pub1% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
tlsse -c -H #R00%priv1% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
tlsse -c -H #s00%V32% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
REM
PAUSE

```

```

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 1018 ms
ClientSession with key1.com opened in 1018 ms

```

```

Send: c00
Recv: OK
in 359 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 978 ms
ClientSession with key1.com opened in 978 ms

```

Send:
 P000495C60DCFF9636B7603BD4C52B4048C260F9B87E3975F7599A072FCBD25B417ADDEACBD972DFD55771E459CB50
 C10079C0AEC0136F822871BC860D771DC554F6C
 Recv: OK
 in 437 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 963 ms
 ClientSession with key1.com opened in 963 ms

Send: R00484B1E61536050B59505943FFED7C8A6A5639B2FF7B42600F889B470E67BF89F
 Recv: OK
 in 406 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 964 ms
 ClientSession with key1.com opened in 964 ms

Send: s001234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
 Recv:
 3045022068413DE51D1FB599905DC480BBA016AE91A411B474AC083627E372054A99EB99022100FD2E91CD69F43A5E
 2ED1A5C22E6F7D85AC22AB8B7CB69ECA944D81F62ADF81F2
 in 538 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 958 ms
 ClientSession with key1.com opened in 958 ms

Send: C00
 Recv: OK
 in 360 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 921 ms
 ClientSession with key1.com opened in 921 ms

Send:
 P000411E4E60C0A2461F3A94FC2F7ACBB6C527855BD08C90B91320AE796178628531033B6C4FCC6C04DD8B278D9723
 8B6C34501FC120D461A16EBB35756D3CFDD43D2
 Recv: OK
 in 453 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 961 ms
 ClientSession with key1.com opened in 961 ms

Send: R00435B6E516AB693CA54267D5BB50956D9C2F54F90449F16D0034A3B2D64EEEF10
 Recv: OK
 in 407 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com
 ConnectTo: key1.com@127.0.0.1:8888
 Using Software Identity Module
 TLS1.3 Client key1.com is connected in 936 ms
 ClientSession with key1.com opened in 936 ms

Send: s001234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
 Recv:
 30450221008E41B005301CF611FAC865EB51CAD8DD6A529FE31D1654AA04CB0DA4B643853602207BE9A378F478EE67
 A17E7FE0CB78DFCE1B587CDD766E1D478A50893DFFD96170
 in 531 ms
 Send: ?02
 Recv: TLS1.3 Disconnected from key1.com

2.9 test_set_rsakey_sign

```
echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
set nb=001
SET V32=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V64=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
set
P=A6DBBA1977E559876A3F42A1AFFD57AA1F402494E22BA17523E4F2729D3145DFBCDE29431A774C30DE6D09CD8DF8
299F1355CFEEC9A3824711C1F3E4B3FFCE02668DEF4CA5085873E10C59C2E68074C208A04297CE081F56679E092319
70D303A56238F7ABFOCB2E09B8BF9A51346EEE3D90A3E2346B83A1D3ED00FFCD36AF6F
set
Q=FE9644F0A6D0EF464238F38B53AF907474C5B12FE637E0095851F17AB1FEA9BD9C745AE43BAEB0664969C2E5B091
21DC07D72235679CE7B36BB50AF74B2258724CE5D4E15F7EAE9D7A21F398EF1634606504B842A6076D6E0CD9B04A3E
0116063EBC3F949324849B2409CD428D2D68959CA4C231D76FC16BB6AAA7EB35DEA9B7
set
DP1=8288C6F001B1F4F651122848616BAA5736977186DFE10E309BE21221C403B60C30AF044CB0B9BF6B43B6BDC287
5C2B037D96D65292804DFCD7518A41D2E73DFCF6C9F6F958EA4C786B08C4780AE757702C5B637D6AFB43B8CEDAFD7E
EDFA2BBA607994E37D29A569E0C51D6FD63354CD2070B524D3A98092EB3E3057EDECE815
set
DQ1=1AD2C856B6ACC2D91B3716E9F21A48550CCC9B7B997CFF9CDC725DC7F8A3054D5603A863EDA9DD2A19C80BE1D9
75A124A6AAD30EDF68D7BCBEB7677D60AC30B77244B0E642027DF157A1E62191E73C0380013C4CC1637CF320657B3E
8743E4072A6AF923F550DEABDDE32E7E3993B4363737E0C21199D3BE3E93C27774C88E07
set
PQ=7EE0068F1F40563006B86799780B5B3E0ADF9243DA2A51FE854CF8032C5E1DE45EE290757E68D7F5FBFEDD53E7B
357C435DA908D3C5250F5842BD3178DE39C063D1B0E01B094537FA5A5E8AB8E0ABB8E8E996A517690DCE42F0E0C524
5AD052E880B6BCEAE6B6E9DB4C387280D2C7EAA826F1BA91CC5370D46E138E5C91860D5
set SIGN=1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF
set
REF=403B383E773512176786574CE741AD51133E9AC5866E7CFE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD
1E04ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03
247C3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DE
D1F1241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D453
172975101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC54A512CFD1581
0E6BAF17525865F3D0D8771522235F887C7BD7503C5CB8
set
MOD=A5EFF4538809B418C9FB7D6EFC8E6528CFA3DAF500B992D7F8CED705933D3E230590DD54C2E2A5BCE4C1F0F48F
716D95B43A0E9AFB6D4F3E1CF4CB8F2C7D9D10E27104A89817419296E93D6FF1D6B9900CC8D7C82D5653C446762C6B
FC4B470AF36D6B997C9D272D0F62AB698997F4433B5226C6E804B13202108ECADF8CB9253B35C96213C75B54E696DA2
E8A08610698D3D4A702DEE6DE5BA450DD82BD5120C33732858EF5F0CFF7CEFA571167EFFDE517EB8F48E1A477582A1
F3DCE5A6BDE2E5FFF7569BB8B29223ADE10BF6D0D1C4B1AECB42EDB5D9B4FE5C02F1CB7E2F3A346D11E29D6DB48D3C
BEF621F83525296EE3B29D102FB9703EBF50F7E929AF59
set EXP=010001
REM
tlsse -c -H #c20 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #R2000%P% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #R2001%Q% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #R2002%DP1% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #R2003%DQ1% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #R2004%PQ% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #P2000%MOD% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #P2001%EXP% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #s20%SIGN% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
echo Refx: %REF%
REM
tlsse -c -H @%nb%#se0%V64%%V32%%V32% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
REM
```

PAUSE

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 1052 ms
ClientSession with key1.com opened in 1052 ms

Send: c20
Recv: OK
in 296 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 917 ms
ClientSession with key1.com opened in 917 ms

Send:
R2000A6DBBA1977E559876A3F42A1AFFD57AA1F402494E22BA17523E4F2729D3145DFBCDE29431A774C30DE6D09CD8
DF8299F1355CFEEC9A3824711C1F3E4B3FFCE02668DEF4CA5085873E10C59C2E68074C208A04297CE081F56679E092
31970D303A56238F7ABF0CB2E09B8BF9A51346EEE3D90A3E2346B83A1D3ED00FFCD36AF6F
Recv: OK
in 609 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 957 ms
ClientSession with key1.com opened in 957 ms

Send:
R2001FE9644F0A6D0EF464238F38B53AF907474C5B12FE637E0095851F17AB1FEA9BD9C745AE43BAEB0664969C2E5B
09121DC07D72235679CE7B36BB50AF74B2258724CE5D4E15F7EAE9D7A21F398EF1634606504B842A6076D6E0CD9B04
A3E0116063EBC3F949324849B2409CD428D2D68959CA4C231D76FC16BB6AAA7EB35DEA9B7
Recv: OK
in 625 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 962 ms
ClientSession with key1.com opened in 962 ms

Send:
R20028288C6F001B1F4F651122848616BAA5736977186DFE10E309BE21221C403B60C30AF044CB0B9BF6B43B6BDC28
75C2B037D96D65292804DFCD7518A41D2E73DFCF6C9F6F958EA4C786B08C4780AE757702C5B637D6AFB43B8CEDAFD7
EEDFA2BBA607994E37D29A569E0C51D6FD63354CD2070B524D3A98092EB3E3057EDECE815
Recv: OK
in 625 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 958 ms
ClientSession with key1.com opened in 958 ms

Send:
R20031AD2C856B6ACC2D91B3716E9F21A48550CCC9B7B997CFF9CDC725DC7F8A3054D5603A863EDA9DD2A19C80BE1D
975A124A6AAD30EDF68D7BCBEB7677D60AC30B77244B0E642027DF157A1E62191E73C0380013C4CC1637CF320657B3
E8743E4072A6AF923F550DEABDDE32E7E3993B4363737E0C21199D3BE3E93C27774C88E07
Recv: OK
in 610 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 927 ms
ClientSession with key1.com opened in 927 ms

Send:
R20047EE0068F1F40563006B86799780B5B3E0ADF9243DA2A51FE854CF8032C5E1DE45EE290757E68D7F5FBFEDD53E
7B357C435DA908D3C5250F5842BD3178DE39C063D1B0E01B094537FA5A5E8AB8E0ABB8E8E996A517690DCE42F0E0C5
245AD052E880B6BCEAE6B6E9DB4C387280D2C7EAA826F1BA91CC5370D46E138E5C91860D5
Recv: OK
in 611 ms
Send: ?02

Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 956 ms
ClientSession with key1.com opened in 956 ms

Send:
P2000A5EFF4538809B418C9FB7D6EFC8E6528CFA3DAF500B992D7F8CED705933D3E230590DD54C2E2A5BCE4C1F0F48
F716D95B43A0E9AFB6D4F3E1CF4CB8F2C7D9D10E27104A89817419296E93D6FF1D6B9900CC8D7C82D5653C446762C6
BFC4B470AF36D6B997C9D272D0F62AB698997F4433B5226C6E804B13202108ECAFB8CB9253B35C96213C75B54E696DA
2E8A08610698D3D4A702DEE6DE5BA450DD82BD5120C33732858EF5F0CFF7CEFA571167EFFDE517EB8F48E1A477582A
1F3DCE5A6BDE2E5FFF7569BB8B29223ADE10BF6D0D1C4B1AECB42EDB5D9B4FE5C02F1CB7E2F3A346D11E29D6DB48D3
CBEF621F83525296EE3B29D102FB9703EBF50F7E929AF59

Recv: OK
in 906 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 943 ms
ClientSession with key1.com opened in 943 ms

Send: P2001010001
Recv: OK
in 297 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 939 ms
ClientSession with key1.com opened in 939 ms

Send: s201234567890ABCDEF1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF
Recv:
403B383E773512176786574CE741AD51133E9AC5866E7CFE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD1E04
ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C
3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DED1F1
241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D4531729
75101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC54A512CFD15810E6B
AF17525865F3D0D8771522235F887C7BD7503C5CB8
in 922 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
Refx:
403B383E773512176786574CE741AD51133E9AC5866E7CFE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD1E04
ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C
3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DED1F1
241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D4531729
75101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC54A512CFD15810E6B
AF17525865F3D0D8771522235F887C7BD7503C5CB8
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 930 ms
ClientSession with key1.com opened in 930 ms

Send:
#se01234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
Recv:
403B383E773512176786574CE741AD51133E9AC5866E7CFE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD1E04
ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C
3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DED1F1
241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D4531729
75101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC54A512CFD15810E6B
AF17525865F3D0D8771522235F887C7BD7503C5CB8403B383E773512176786574CE741AD51133E9AC5866E7CFE7A3D
BE47453E0095DC0C505DC004FA8445B2AA96DD1E04ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508A
A55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4
CD05F45ED52C0001B961D2C706B7F3710863DED1F1241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659
A47415724BC6311E859CDFACD7A0E6FDA9D453172975101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111
292A38C06ED841B7F5F6182FC54A512CFD15810E6BAF17525865F3D0D8771522235F887C7BD7503C5CB8403B383E77
3512176786574CE741AD51133E9AC5866E7CFE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD1E04ED1DED4CC6
81B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C3F0490A8D3
9E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DED1F1241399C08C
09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D453172975101270EB
AB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC54A512CFD15810E6BAF17525865
F3D0D8771522235F887C7BD7503C5CB8403B383E773512176786574CE741AD51133E9AC5866E7CFE7A3DBE47453E00

```

95DC0C505DC004FA8445B2AA96DD1E04ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECF593A3F508AA55F37E4B1
F91DEECE3EAC1019E1D542CDFE03247C3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED5
2C0001B961D2C706B7F3710863DED1F1241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724B
C6311E859CDFACD7A0E6FDA9D453172975101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06E
D841B7F5F6182FC54A512CFD15810E6BAF17525865F3D0D8771522235F887C7BD7503C5CB8
in 1495 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com

```

2.10 test_pipe_sign

```

@echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
set nb=001
SET V32=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V64=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef
SET
V128=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V256=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345
67890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123
4567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1
234567890abcdef1234567890abcdef1234567890abcdef
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
set nb=001
tlssse -c -H #C00 -H #G00 -H @%nb%#sc0%V256% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l
TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
tlssse -c -H #c20 -H #g20 -H @%nb%#se0%V64%#V32%#V32% -H #?02 -S %SEN% -s -p %PORT% -h %IP%
-l TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
REM
PAUSE

```

```

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 891 ms
ClientSession with key1.com opened in 891 ms

```

```

Send: C00
Recv: OK
in 329 ms
Send: G00
Recv: OK
in 140 ms
Send:
#sc01234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456
7890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234
567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12
34567890abcdef1234567890abcdef1234567890abcdef
Recv:
3045022100CF98EB9F7B996122F903F53DA0ECCDE3DA004C8B8291D293073B07B61BA3B3D7022034FA8C3906910411
107DC47B08589DD15885039306FCACCD23FC72482C97523045022100829E7FAA3E3D4DB5BBF5BCAC41C9C0013120
4A2D60BEEEE199459DB06EB843B2B02201699D0F3F6D888CC279CB525CEDADC74BE37DB91317B77824BE476075FA4B5
783046022100F63A8E9221DEBDBBF2D35D7F6C1CFA9792390FF3E7DC879F7F097B81427D69EB022100DD233FDCADB3
5684DC462916D284B73A1071A6F75C8904704B85E79019893B903045022018F1CE62C77E101FA0012CCF76892DAAB9
0A36CE2A73FE80EC9302610265ADDC0221008453280F6BEF8667E3CD85F86091AE6905B904ADA9386EC705E4CFBB6C
7449723046022100AA400E40491EE11105F3EC61ADBADF083607FA3C8640BAE5CAC65B343CAA3F40022100D92793EB
D64048EC095EF553A27FCC6F2921BB28102B62DB4C0F5741359413E530450220683411627797C81AFB043A9C572529
0EE9721440785EF736D8B4E109CDE3943A022100861350172347E2E2D3F76C6DBB90C3D2C09C773BBAF014043121A
005FEA53DE3046022100FE09FC563EEED77249FA8AA05471E5BDF1D23D1D28424A9AF9AFB6D2A0D145C1022100F364
6375008AA4F3142982096E29B787439EDDBFD25544AA02503C4DD726BF71304602210096A6B0711B45DBE6209F48D9

```

```

4E20ACD2BDCFD5F8E3225DA99DE35A1070F1C63F022100C1E97F5FB02D99DD747507D12513E3B4A60D833393ED716F
B780AE30E2C30678
in 1001 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 798 ms
ClientSession with key1.com opened in 798 ms

Send: c20
Recv: OK
in 283 ms
Send: g20
Recv: OK
in 1550 ms
Send:
#se01234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv:
383FFF4F4F336A9BB1A02A7E96D645BB9E94739EE9AAE4A0840E3A367D0748830BA1D063DCA78F59533C506743909E
14FB421801BB6DC5C3179D88657FF4409A6A75A648A509DDECE558BF65963B385A9A813D73EDCBAA56B6414897B026
CFD0B3CF048A30264D3541B82759C556E0CA7C635471DB3527D2CDF159AD28E3B67B07E7785F846DA9B0DD8D321B97
B746509A923E9BC7CD2E8B8B6B1A63BEEE49D838C3E149F827AE26D66BDDC0A32A31AF7788E08A19D3DC2A48DC7F1C
41B73E36AA3C626D3985C3AEFF88CE3F83BBE8233215B310DDC8050840F3C119A6275A29A209BEEB0126454BCD2585
97962CD02458DA77183A552CCBAC647B1F7B48FD61383FFF4F4F336A9BB1A02A7E96D645BB9E94739EE9AAE4A0840E
3A367D0748830BA1D063DCA78F59533C506743909E14FB421801BB6DC5C3179D88657FF4409A6A75A648A509DDECE5
58BF65963B385A9A813D73EDCBAA56B6414897B026CFD0B3CF048A30264D3541B82759C556E0CA7C635471DB3527D2
CDF159AD28E3B67B07E7785F846DA9B0DD8D321B97B746509A923E9BC7CD2E8B8B6B1A63BEEE49D838C3E149F827AE
26D66BDDC0A32A31AF7788E08A19D3DC2A48DC7F1C41B73E36AA3C626D3985C3AEFF88CE3F83BBE8233215B310DDC8
050840F3C119A6275A29A209BEEB0126454BCD258597962CD02458DA77183A552CCBAC647B1F7B48FD61383FFF4F4F
336A9BB1A02A7E96D645BB9E94739EE9AAE4A0840E3A367D0748830BA1D063DCA78F59533C506743909E14FB421801
BB6DC5C3179D88657FF4409A6A75A648A509DDECE558BF65963B385A9A813D73EDCBAA56B6414897B026CFD0B3CF04
8A30264D3541B82759C556E0CA7C635471DB3527D2CDF159AD28E3B67B07E7785F846DA9B0DD8D321B97B746509A92
3E9BC7CD2E8B8B6B1A63BEEE49D838C3E149F827AE26D66BDDC0A32A31AF7788E08A19D3DC2A48DC7F1C41B73E36AA
3C626D3985C3AEFF88CE3F83BBE8233215B310DDC8050840F3C119A6275A29A209BEEB0126454BCD258597962CD024
58DA77183A552CCBAC647B1F7B48FD61383FFF4F4F336A9BB1A02A7E96D645BB9E94739EE9AAE4A0840E3A367D0748
830BA1D063DCA78F59533C506743909E14FB421801BB6DC5C3179D88657FF4409A6A75A648A509DDECE558BF65963B
385A9A813D73EDCBAA56B6414897B026CFD0B3CF048A30264D3541B82759C556E0CA7C635471DB3527D2CDF159AD28
E3B67B07E7785F846DA9B0DD8D321B97B746509A923E9BC7CD2E8B8B6B1A63BEEE49D838C3E149F827AE26D66BDDC0
A32A31AF7788E08A19D3DC2A48DC7F1C41B73E36AA3C626D3985C3AEFF88CE3F83BBE8233215B310DDC8050840F3C1
19A6275A29A209BEEB0126454BCD258597962CD02458DA77183A552CCBAC647B1F7B48FD61
in 1213 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com

```

2.11 test_aes

```

@echo off
set IP=127.0.0.1
set PORT=8888
set SEN=key1.com
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
REM https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-38a.pdf
REM F.1 ECB Example Vectors
REM F.1.1 ECB-AES128.Encrypt
REM Key 2b7e151628aed2a6abf7158809cf4f3c
REM Plaintext      6bc1bee22e409f96e93d7e117393172a
REM Ciphertext     3ad77bb40d7a3660a89ecaf32466ef97
REM Plaintext      ae2d8a571e03ac9c9eb76fac45af8e51
REM Ciphertext     f5d3d58503b9699de785895a96fdbaaaf
REM Plaintext      30c81c46a35ce411e5fbc1191a0a52ef
REM Ciphertext     43b1cd7f598ece23881b00e3ed030688
REM Plaintext      f69f2445df4f9b17ad2b417be66c3710
REM Ciphertext     7b0c785e27e8ad3f8223207104725dd4
REM
REM
set nb=001
set
H256=6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117
393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e1
17393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7

```

[illegible]

```
Send: t002b7e151628aed2a6abf7158809cf4f3c
Recv: OK
in 317 ms
Send: A006bc1bee22e409f96e93d7e117393172a
Recv: 3AD77BB40D7A3660A89ECAf32466EF97
in 154 ms
Send: a003ad77bb40d7a3660a89ecaf32466ef97
Recv: 6BC1BEE22E409F96E93D7E117393172A
in 141 ms
Send: t012b7e151628aed2a6abf7158809cf4f3c
Recv: OK
in 110 ms
Send: A01ae2d8a571e03ac9c9eb76fac45af8e51
Recv: F5D3D58503B9699DE785895A96FDBAAF
in 140 ms
Send: a01f5d3d58503b9699de785895a96fdbaa
Recv: AE2D8A571E03AC9C9EB76FAC45AF8E51
in 142 ms
Send: t022b7e151628aed2a6abf7158809cf4f3c
Recv: OK
in 110 ms
Send: A0230c81c46a35ce411e5fbc1191a0a52ef
Recv: 43B1CD7F598ECE23881B00E3ED030688
in 141 ms
Send: a0243b1cd7f598ece23881b00e3ed030688
```

```
Send:  
A006bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e11739  
3172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117  
393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e1  
17393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e  
e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d  
7e117393172a6bc1bee22e409f96e93d7e117393172a  
Recv:  
3AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF  
973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466  
EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF324  
66EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF3  
2466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECA  
F32466EF973AD77BB40D7A3660A89ECAAF32466EF97  
in 1401 ms  
Send: ?02  
Recv: TLS1.3 Disconnected from keyl.com  
ConnectTo: keyl.com@127.0.0.1:8888  
Using Software Identity Module  
TLS1.3 Client keyl.com is connected in 815 ms  
ClientSession with keyl.com opened in 815 ms  
  
Send: #Ac06bc1bee22e409f96e93d7e117393172a  
Recv: 3AD77BB40D7A3660A89ECAAF32466EF97  
in 317 ms  
Send: #ac3ad77bb40d7a3660a89ecaf32466ef97  
Recv: TLS1.3 Disconnected from keyl.com  
ConnectTo: keyl.com@127.0.0.1:8888  
Using Software Identity Module  
TLS1.3 Client keyl.com is connected in 805 ms  
ClientSession with keyl.com opened in 805 ms  
  
Send:  
#ac06bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e1173  
93172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e11  
7393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e  
117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d  
7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93  
3d7e117393172a6bc1bee22e409f96e93d7e117393172a  
Recv:  
500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C38F085  
F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C38F0  
85F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C38  
F085F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C  
38F085F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B667C38F085F1500594E20D6D6C8C716B66  
7C38F085F1500594E20D6D6C8C716B667C38F085F1  
in 742 ms  
Send: ?02  
Recv: TLS1.3 Disconnected from keyl.com  
ConnectTo: keyl.com@127.0.0.1:8888  
Using Software Identity Module  
TLS1.3 Client keyl.com is connected in 807 ms  
ClientSession with keyl.com opened in 807 ms  
  
Send:  
#Ac06bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e1173  
93172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e11  
7393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e  
117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d  
7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93
```

```
7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e9  
3d7e117393172a6bc1bee22e409f96e93d7e117393172a
```

[illegible]

```
in 1185 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 811 ms
ClientSession with key1.com opened in 811 ms
```

Recv:
3AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF
973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466
EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF324
66EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF3

```
2466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECA
F32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89E
CAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A8
9ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660
A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A36
60A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A
3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D
7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB4
0D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77B
B40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD7
7BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973A
D77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF97
3AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF
973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466
EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF324
66EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF3
2466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECA
F32466EF973AD77BB40D7A3660A89ECAAF32466EF973AD77BB40D7A3660A89ECAAF32466EF97
in 2074 ms
```

```
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
```

2.12 test_hmac

```
@echo off
set IP=127.0.0.1
set PORT=8888
set SEN=key1.com
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
set nb=001
set
H256=6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117
393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e1
17393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e
e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93
d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e93d7e117393172a6bc1bee22e409f96e
93d7e117393172a6bc1bee22e409f96e93d7e117393172a
SET V32=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V64=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef
SET
V128=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V256=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345
67890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123
4567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1
234567890abcdef1234567890abcdef1234567890abcdef
REM
tlsse -c -H #t00%V32% -H #h00%V32% -H #H00%V32% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l
TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
REM
tlsse -c -H #h40%V256% -H #H40%V128% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
REM
tlsse -c -H ##hc0%V256%V256% -H ##Hc0%V256%V256% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l
TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
REM
tlsse -c -H @%nb%#hc0%V256%V256% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H @%nb%#Hc0%V256%V256% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
REM
REM
PAUSE

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
```

TLS1.3 Client key1.com is connected in 899 ms
ClientSession with key1.com opened in 899 ms

Send: t001234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: OK
in 362 ms
Send: h001234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: DCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42E
in 214 ms
Send: H001234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv:
B91821A06051C98F2558360FA984A7FF5558B367FF195E3E5EBCE7DB4CB9E94AC4A6DF4CB4D7FE29D11C556AB316BE
F2A676827A0F780926BACFF9CA595E402C
in 253 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 828 ms
ClientSession with key1.com opened in 828 ms

Send:
h401234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890a
bcedef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456
7890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345
4567890abcdef1234567890abcdef1234567890abcdef
Recv:
DCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE02B05
5064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284
A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE0
2B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E
4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E2
5DE02B055064BABCFF687E6F8BF583868E4284A42E
in 1482 ms
Send:
H401234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890a
bcedef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv:
B91821A06051C98F2558360FA984A7FF5558B367FF195E3E5EBCE7DB4CB9E94AC4A6DF4CB4D7FE29D11C556AB316BE
F2A676827A0F780926BACFF9CA595E402CB91821A06051C98F2558360FA984A7FF5558B367FF195E3E5EBCE7DB4CB9
E94AC4A6DF4CB4D7FE29D11C556AB316BEF2A676827A0F780926BACFF9CA595E402C469CCE80064300ECB49882A7B1
EA22844E357D36A9A75CE2E0B8B340C1D7F4CE8F32D87351FCAED74FBF7D3A554A5F53A841F137173D137EEB9EF826
EF54A30D469CCE80064300ECB49882A7B1EA22844E357D36A9A75CE2E0B8B340C1D7F4CE8F32D87351FCAED74FBF7D
3A554A5F53A841F137173D137EEB9EF826EF54A30D
in 916 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 810 ms
ClientSession with key1.com opened in 810 ms

Send:
#hc01234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456
7890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234
567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12
34567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcd
ef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890ab
cdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv:
DCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE02B05
5064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284
A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE0
2B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E
4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E2
5DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583
868E4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA0
45E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8B
F583868E4284A42EDCBC5A8291C33FFCA045E25DE02B055064BABCFF687E6F8BF583868E4284A42EDCBC5A8291C33F


```
Send:
#Hc01234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456
7890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234
567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12
34567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcd
ef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890ab
cdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv:
B91821A06051C98F2558360FA984A7FF5558B367FF195E3E5EBCE7DB4CB9E94AC4A6DF4CB4D7FE29D11C556AB316BE
F2A676827A0F780926BACFF9CA595E402CB91821A06051C98F2558360FA984A7FF5558B367FF195E3E5EBCE7DB4CB9
E94AC4A6DF4CB4D7FE29D11C556AB316BEF2A676827A0F780926BACFF9CA595E402CB91821A06051C98F2558360FA9
84A7FF5558B367FF195E3E5EBCE7DB4CB9E94AC4A6DF4CB4D7FE29D11C556AB316BEF2A676827A0F780926BACFF9CA
595E402CB91821A06051C98F2558360FA984A7FF5558B367FF195E3E5EBCE7DB4CB9E94AC4A6DF4CB4D7FE29D11C55
6AB316BEF2A676827A0F780926BACFF9CA595E402CB91821A06051C98F2558360FA984A7FF5558B367FF195E3E5EBC
E7DB4CB9E94AC4A6DF4CB4D7FE29D11C556AB316BEF2A676827A0F780926BACFF9CA595E402CB91821A06051C98F25
58360FA984A7FF5558B367FF195E3E5EBCE7DB4CB9E94AC4A6DF4CB4D7FE29D11C556AB316BEF2A676827A0F780926
BACFF9CA595E402CB91821A06051C98F2558360FA984A7FF5558B367FF195E3E5EBCE7DB4CB9E94AC4A6DF4CB4D7FE
29D11C556AB316BEF2A676827A0F780926BACFF9CA595E402CB91821A06051C98F2558360FA984A7FF5558B367FF19
5E3E5EBCE7DB4CB9E94AC4A6DF4CB4D7FE29D11C556AB316BEF2A676827A0F780926BACFF9CA595E402C891145208B
6466153B81AE233FC82277A843D96D660FB6C92A877C4E8757F84BE6B6E1A4D2A04EEC8128EF842B060F70CD08063B
694E82E597887AE5CE3EEA9ACAE3EB24733DD1237B42CA2958D772BE7770E9AF8CFA3A794D0E42E3E6BA7070950011
20255F608CD91B16B7ED4D1E6A958C7819A4E3CE738A3E4E1D9CADF0E3CAE3EB24733DD1237B42CA2958D772BE7770
E9AF8CFA3A794D0E42E3E6BA707095001120255F608CD91B16B7ED4D1E6A958C7819A4E3CE738A3E4E1D9CADF0E3CA
E3EB24733DD1237B42CA2958D772BE7770E9AF8CFA3A794D0E42E3E6BA707095001120255F608CD91B16B7ED4D1E6A
958C7819A4E3CE738A3E4E1D9CADF0E3CAE3EB24733DD1237B42CA2958D772BE7770E9AF8CFA3A794D0E42E3E6BA70
7095001120255F608CD91B16B7ED4D1E6A958C7819A4E3CE738A3E4E1D9CADF0E3CAE3EB24733DD1237B42CA2958D7
72BE7770E9AF8CFA3A794D0E42E3E6BA707095001120255F608CD91B16B7ED4D1E6A958C7819A4E3CE738A3E4E1D9C
ADF0E3CAE3EB24733DD1237B42CA2958D772BE7770E9AF8CFA3A794D0E42E3E6BA707095001120255F608CD91B16B7
ED4D1E6A958C7819A4E3CE738A3E4E1D9CADF0E3CAE3EB24733DD1237B42CA2958D772BE7770E9AF8CFA3A794D0E42
E3E6BA707095001120255F608CD91B16B7ED4D1E6A958C7819A4E3CE738A3E4E1D9CADF0E3
in 1771 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
```

2.13 test_bip32

```
@echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
set
seed=f0acfacc9279b18275f6923a1f2d2c2be622e39e7bda9ed6d8a783f200b4a011793df8cde50014b5d87169712
21dc7c63e6ff4e22128efa314b4c32fc07e76ae
REM //m/0'/1'/2'/3'
set path=800000008000000018000000280000003
REM
tlsse -c -H #c38 -H #t38%seed% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #b00%path% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
tlsse -c -H #p00 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
echo Refx: 0282e66e0390200bf7264f7a3c02ed24767baeeda9d9205c171577aaf155d3cb32
tlsse -c -H #r00 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256
-H psk%PSK% -H identity%id%
REM
PAUSE
```

```
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 908 ms
ClientSession with key1.com opened in 908 ms
```

```

Send: c38
Recv: OK
in 283 ms
Send:
t38f0acfacf9279b18275f6923a1f2d2c2be622e39e7bda9ed6d8a783f200b4a011793df8cde50014b5d8716971221
dc7c63e6ff4e22128efa314b4c32fc07e76ae
Recv: OK
in 251 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 826 ms
ClientSession with key1.com opened in 826 ms

Send: b0080000000800000018000000280000003
Recv: OK
in 534 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 829 ms
ClientSession with key1.com opened in 829 ms

Send: p00
Recv:
0482E66E0390200BF7264F7A3C02ED24767BAEEDA9D9205C171577AAF155D3CB32C28EA997BD8905F8F7446269C88A
849EDB5CB83D3B17009560FCAFA8199C466E
in 392 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
Refx: 0282e66e0390200bf7264f7a3c02ed24767baeeda9d9205c171577aaf155d3cb32
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 813 ms
ClientSession with key1.com opened in 813 ms

Send: r00
Recv: E4886AA0DCDAD6629EA14A84FBA0DA09352426E12E752B085F3372E7D0A28125
in 330 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com

```

2.14 test_rw_1280

```

@echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
SET V32=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V64=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef
SET
V128=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V127=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcd
SET
V256=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345
67890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123
4567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1
234567890abcdef1234567890abcdef1234567890abcdef
REM
:loop

```

```
tlssse -c -H #Z001234 -H #I00 -H ##I00 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-  
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%  
tlssse -c -H ##Z00313233343536 -H ##Z80313233343536 -H ##I00 -H #I80 -H #?02 -S %SEN% -s -p  
%PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%  
tlssse -c -H #Z00%V256%%V256%%V128% -H #I00 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-  
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%  
tlssse -c -H ##Z00%V256%%V256%%V128%%V256%%V256%%V128% -H ##I00 -H #?02 -S %SEN% -s -p  
%PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%  
tlssse -c -H #Z01%V256%%V256%%V128% -H #I01 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-  
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%  
tlssse -c -H #Z02%V256%%V256%%V128% -H #I02 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-  
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%  
tlssse -c -H #Z03%V256%%V256%%V128% -H #I03 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-  
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%  
REM  
PAUSE
```

```
ConnectTo: key1.com@127.0.0.1:8888  
Using Software Identity Module  
TLS1.3 Client key1.com is connected in 902 ms  
ClientSession with key1.com opened in 902 ms
```

```
Send: Z001234  
Recv: OK  
in 297 ms  
Send: I00  
Recv: 1234  
in 66 ms  
Send: #I00  
Recv: 31323334  
in 62 ms  
Send: ?02  
Recv: TLS1.3 Disconnected from key1.com  
ConnectTo: key1.com@127.0.0.1:8888  
Using Software Identity Module  
TLS1.3 Client key1.com is connected in 807 ms  
ClientSession with key1.com opened in 807 ms
```

```
Send: #Z00313233343536  
Recv: 4F4B  
in 299 ms  
Send: #Z80313233343536  
Recv: 4F4B  
in 63 ms  
Send: #I00  
Recv: 313233343536  
in 63 ms  
Send: I80  
Recv: 123456  
in 63 ms  
Send: ?02  
Recv: TLS1.3 Disconnected from key1.com  
ConnectTo: key1.com@127.0.0.1:8888  
Using Software Identity Module  
TLS1.3 Client key1.com is connected in 821 ms  
ClientSession with key1.com opened in 821 ms
```

```
Send:  
Z001234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890a  
bcedef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789  
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567  
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345  
67890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123  
4567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1  
234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcde  
f1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abc  
def1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890a  
bcedef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789  
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567  
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345  
67890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123  
4567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef  
Recv: OK  
in 1361 ms  
Send: I00  
Recv:  
1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcd
```

```
ef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890ab
cdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456
7890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234
567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12
34567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcd
ef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890ab
cdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345678
90abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456
7890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
in 1102 ms
Send: ?02
Recv: TLS1.3 Disconnected from keyl.com
ConnectTo: keyl.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client keyl.com is connected in 815 ms
ClientSession with keyl.com opened in 815 ms
```



```

SET
V127=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcd
SET
V256=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345
67890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123
4567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1
234567890abcdef1234567890abcdef1234567890abcdef
REM
tlsse -c -H #Z04%V32% -H #I04 -H ##I04 -H #Z04 -H #I04 -H ##I04 -H #?02 -S %SEN% -s -p
%PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #Z05%V32% -H #I05 -H #Z05 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #Z06%V32% -H #I06 -H #Z06 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #Z07%V32% -H #I07 -H #Z07 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #Z08%V32% -H #I3F -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-
SHA256 -H psk%PSK% -H identity%id%
REM
PAUSE

```

```

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 899 ms
ClientSession with key1.com opened in 899 ms

```

```

Send: Z041234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: OK
in 334 ms
Send: I04
Recv: 1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
in 125 ms
Send: #I04
Recv:
3132333435363738393061626364656631323334353637383930616263646566313233343536373839306162636465
6631323334353637383930616263646566
in 110 ms
Send: Z04
Recv: OK
in 62 ms
Send: I04
Recv:
in 63 ms
Send: #I04
Recv:
in 73 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 822 ms
ClientSession with key1.com opened in 822 ms

```

```

Send: Z051234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: OK
in 330 ms
Send: I05
Recv: 1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
in 102 ms
Send: Z05
Recv: OK
in 63 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 815 ms
ClientSession with key1.com opened in 815 ms

```

```

Send: Z061234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: OK
in 342 ms

```

```

Send: I06
Recv: 1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
in 109 ms
Send: Z06
Recv: OK
in 63 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 819 ms
ClientSession with key1.com opened in 819 ms

Send: Z071234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: OK
in 331 ms
Send: I07
Recv: 1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
in 116 ms
Send: Z07
Recv: OK
in 65 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 824 ms
ClientSession with key1.com opened in 824 ms

Send: Z081234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: ERROR
in 323 ms
Send: I3F
Recv: TLS1.3 Disconnected from key1.com

```

2.16 test_derive

```

@echo off
set IP=127.0.0.1
set PORT=8888
set SEN=key1.com
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client identity
SET V32=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
REM
tlsse -c -H #?A2%PSK% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
REM
tlsse -c -H #?D0%V32% -H #?D1%V32% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
PAUSE

```

```

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 920 ms
ClientSession with key1.com opened in 920 ms

Send: ?A20102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
Recv: 0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
in 461 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 825 ms
ClientSession with key1.com opened in 825 ms

Send: ?D01234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: F27A333ECCE8A1A9463514245D4E1F5346F7D43360F30D23E6918BD66FB19AC7
in 439 ms
Send: ?D11234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv: 380E7A39E701820E4E96E4A2E2AB8C7365F5C22332D72EA00323BDE1971071E0

```

```
in 192 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
```

2.17 test_export_ecc

```
@echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
set key=1234567890ABCDEF1234567890ABCDEF
set type=00
set index=10
set param=80
set keyid=E1E2E3E4
set
pub1=0411E4E60C0A2461F3A94FC2F7ACBB6C527855BD08C90B91320AE796178628531033B6C4FCC6C04DD8B278D97
238B6C34501FC120D461A16EBB35756D3CFDD43D2
set priv1=435B6E516AB693CA54267D5BB50956D9C2F54F90449F16D0034A3B2D64EEEF10
REM
1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF
EF1234567890ABCDEF1234567890ABCDEF
set
blob0=001000E1E2E3E4EF73F42F9470C440D29221FFBD9B9ECCC66CB7C26F9690700BF16DC85F7F4393727331DEAB
3A127AADE170C7E93377B484DCC4BBA210608EA006D7A2FEA77EC5BA7451F51A12B683C7DF21C9A16669633D5F849E
0A25B5A91EBC01C863
set
blob1=001080E1E2E3E4C67F9FE01F7BF420030B56ED5C11C67B89BB2299721161BE53B109C60621F30F2139F494E8
312CDD8330389DE8CAA7AC3736EFAA665F1D2534173E09
REM
tlsse -c -H #?30?key% -H #c%index% -H #P00?pub1% -H #R00?priv1% -H #p%index% -H #r%index%
-H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #?33?type%%index%%param%%keyid% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-
AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
REM For ECC PubKey is computed from PrivKey
tlsse -c -H #c%index% -H #p%index% -H #?34?blob1% -H #r%index% -H #?02 -S
%SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
REM
PAUSE

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 916 ms
ClientSession with key1.com opened in 916 ms

Send: ?301234567890ABCDEF1234567890ABCDEF
Recv: OK
in 317 ms
Send: c10
Recv: OK
in 128 ms
Send:
P000411E4E60C0A2461F3A94FC2F7ACBB6C527855BD08C90B91320AE796178628531033B6C4FCC6C04DD8B278D9723
8B6C34501FC120D461A16EBB35756D3CFDD43D2
Recv: OK
in 203 ms
Send: R00435B6E516AB693CA54267D5BB50956D9C2F54F90449F16D0034A3B2D64EEEF10
Recv: OK
in 172 ms
Send: p10
Recv:
0411E4E60C0A2461F3A94FC2F7ACBB6C527855BD08C90B91320AE796178628531033B6C4FCC6C04DD8B278D97238B6
C34501FC120D461A16EBB35756D3CFDD43D2
in 189 ms
Send: r10
Recv: 435B6E516AB693CA54267D5BB50956D9C2F54F90449F16D0034A3B2D64EEEF10
in 125 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
```

```

Using Software Identity Module
TLS1.3 Client key1.com is connected in 818 ms
ClientSession with key1.com opened in 818 ms

Send: ?33001080E1E2E3E4
Recv:
001080E1E2E3E48A73448A85E9F91B804F5804124679379A1F89A6254D8E87C61493A27E060ED3059F17F1550684F4
BE3ABF977FC00FE8C8F5A17565FF0ABE27DDA14D
in 461 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 831 ms
ClientSession with key1.com opened in 831 ms

Send: c10
Recv: OK
in 331 ms
Send: p10
Recv: ERROR
in 79 ms
Send:
?34001080E1E2E3E4C67F9FE01F7BF420030B56ED5C11C67B89BB2299721161BE53B109C60621F30F2139F494E8312
CDD8330389DE8CAA7AC3736EFAA665F1D2534173E09
Recv: OK
in 301 ms
Send: p10
Recv:
0411E4E60C0A2461F3A94FC2F7ACBB6C527855BD08C90B91320AE796178628531033B6C4FCC6C04DD8B278D97238B6
C34501FC120D461A16EBB35756D3CFDD43D2
in 175 ms
Send: r10
Recv: 435B6E516AB693CA54267D5BB50956D9C2F54F90449F16D0034A3B2D64EEEF10
in 111 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com

```

2.18 test_export_rsa

```

@echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
set nb=001
SET V32=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V64=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890
abcdef1234567890abcdef1234567890abcdef
SET
V128=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
SET
V256=1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123456789
0abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567
890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef12345
67890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef123
4567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1
234567890abcdef1234567890abcdef1234567890abcdef
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
set key=1234567890ABCDEF1234567890ABCDEF
set type=00
set keyid=E1E2E3E4
REM
set
P=A6DBBA1977E559876A3F42A1AFFD57AA1F402494E22BA17523E4F2729D3145DFBCDE29431A774C30DE6D09CD8DF8
299F1355CFEEC9A3824711C1F3E4B3FFCE02668DEF4CA5085873E10C59C2E68074C208A04297CE081F56679E092319
70D303A56238F7ABFOCB2E09B8BF9A51346EEE3D90A3E2346B83A1D3ED00FFCD36AF6F
set
Q=FE9644F0A6D0EF464238F38B53AF907474C5B12FE637E0095851F17AB1FEA9BD9C745AE43BAEB0664969C2E5B091

```

```
21DC07D72235679CE7B36BB50AF74B2258724CE5D4E15F7EAE9D7A21F398EF1634606504B842A6076D6E0CD9B04A3E
0116063EBC3F949324849B2409CD428D2D68959CA4C231D76FC16BB6AAA7EB35DEA9B7
set
DP1=8288C6F001B1F4F651122848616BAA5736977186DFE10E309BE21221C403B60C30AF044CB0B9BF6B43B6BDC287
5C2B037D96D65292804DFC7518A41D2E73DFCF6C9F6F958EA4C786B08C4780AE757702C5B637D6AFB43B8CEDAFD7E
EDFA2BBA607994E37D29A569E0C51D6FD63354CD2070B524D3A98092EB3E3057EDECE815
set
DQ1=1AD2C856B6ACC2D91B3716E9F21A48550CCC9B7B997CFF9CDC725DC7F8A3054D5603A863EDA9DD2A19C80BE1D9
75A124A6AAD30EDF68D7BCBEB7677D60AC30B77244B0E642027DF157A1E62191E73C0380013C4CC1637CF320657B3E
8743E4072A6AF923F550DEABDD3E2E7E3993B4363737E0C21199D3BE3E93C27774C88E07
set
PQ=7EE0068F1F40563006B86799780B5B3E0ADF9243DA2A51FE854CF8032C5E1DE45EE290757E68D7F5FBFEDD53E7B
357C435DA908D3C5250F5842BD3178DE39C063D1B0E01B094537FA5A5E8AB8E0ABB8E8E996A517690DCE42F0E0C524
5AD052E880B6BCEAE6B6E9DB4C387280D2C7EAA826F1BA91CC5370D46E138E5C91860D5
set SIGN=1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF
set
REF=403B383E773512176786574CE741AD51133E9AC5866E7CFE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD
1E04ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03
247C3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DE
D1F1241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D453
172975101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC5A4512CFD1581
0E6BAF17525865F3D0D8771522235F887C7BD7503C5CB8
set
MOD=A5EFF4538809B418C9FB7D6EFC8E6528CFA3DAF500B992D7F8CED705933D3E230590DD54C2E2A5BCE4C1F0F48F
716D95B43A0E9AFB6D4F3E1CF4CB8F2C7D9D10E27104A89817419296E93D6FF1D6B9900CC8D7C82D5653C446762C6B
FC4B470AF36D6B997C9D272D0F62AB698997F4433B5226C6E804B13202108ECADF8CB9253B35C96213C75B54E696DA2
E8A08610698D3D4702DEE6DE5BA450DD82BD5120C33732858EF5F0CFF7CEFA571167EFFDE517EB8F48E1A477582A1
F3DCE5A6BDE2E5FFF7569BB8B29223ADE10BF6D0D1C4B1AECB42EDB5D9B4FE5C02F1CB7E2F3A346D11E29D6DB48D3C
BEF621F83525296EE3B29D102FB9703EBF50F7E929AF59
set EXP=010001
REM
set
blob0=002000E1E2E3E4339A818162E9790A1D9BEE70C6A1838D037BD8A4E479653E86A566E9A3BA90462DA19206F6
5572D48BF546396B5C56849596DE63901517C2969FD557C96ABF127CCC304A1C40B19964FA219645087CBCBD7CD32E
816864CAA04C36BE24C31E0DD751F084E21F3E9182FC6E42705009BB9017623CC9DE4507FC4D5FAEBC99FC7B4041F8
CF4C98A11D8F9C680752D9582A86C4EFED0100C9A0FCEE1A0C9DA6A15477311C6A7E97D45B5D447231805C50F29125
7759737FEF13C91DD23210D7880FDAC02255CC661648D1B3998DC45E951328F64D24116A51B1B40A78C422D8A66641
FBF714133B649959CEA8274C5199145836679CD581AF51A014A64C7165D252627699CC94E39E87BF0A022A48FEC2A7
A7E6440BB6B52748BB0A7FFA
set blob1=002001E1E2E3E4295BAF58E4F23CF9525A79792E21E8BC123EE87EB3A56CB692850B782B2DEE
set
blob2=002080E1E2E3E4FD836F4961770D56B04ED86D4C2FBE41B62880613DC84ECB86ACAFFFE5397EBB9D543ECC4F
4F6DF9A5E12C8F1DDF0BDA45E97A460354D38FB81F04A48E5B2199A52E2FAE33BB9F048541CD8C8811EA91ED584996
01559B9E263529534AB87C2DB3A6FD622DADE5B968FFA80B132C6F91AD612C5E39E7F082B524F94E001B260BE0F9
2B803EF00505D78D67D44183A9568BA4FDBFBFBFC1CC01778E2
set
blob3=002081E1E2E3E4A7D12543C97895112E1EAA8A7E1EB9BABF56DE9CCA270D1A83472C54A2C52FEEB6F0760492
1AE095009458561D8F064BF9990082777AEDC53B105A8E6C75D1D44F7CB725975E8441ABDBEC086BE12298A340CD2D
49CC370E88AC964074B934C52BEB1303FB6E75515D0ECD18731F266EFD4DE2A5E2AA7EF983FF94621C77E8AC75301
57AB71ABC1E00703EFB9B6255BAC9D2A09CB53726580824D8A
set
blob4=002082E1E2E3E4E00BE9D6A32F9715D065E9629527F8C6366D494CCB60DD1AC3B666EA9A94CCBA177D9BC7F2
8E2D2DC6E71996F09930197E83C92FC1CE4EA34E1EB2928CFD851CF9324C0A62B541F8A68ED9407CE0B340E1E037C7
E41DB4ECFFBF666DA94CC78776D9138C5DBDECCD5A1783E9D80FB1D49823617369B9C05860D78ADB421FB4FFBB23ED
F91789D3A05C6C547301722B063D1B4CAB5B82F70E6AB3DA6F
set
blob5=002083E1E2E3E4C18D255E9F2C0F4AF9F94D978BA318AFF40C6FFCE0A0EF810FBE40CFC2AF14C6081953C7DF
EE1A1F0099FA16E6D5F47F61BE254C729549D49BA15422C2DF6F0CB3D9E9612380ED4586DCAFF9F92862A40FAF0416
220F9028838960B4F767240E8B932CBF4BAC7D7E0783DDAA7F7F1D2CC8ACADBC0B49BAB17DAC8219438903750F6024
B842BEE47968D189577AD5498698F2D2EC157A691D5E6FB025
set
blob6=002084E1E2E3E4CA9BF77B6F8CCB7A55EF84B408DB6155831DEFF31F36746BD9BBF5CABC1B3354941A425D36
12810D27BD63940530E5E3E0D373C9D2AB08D0E2B248B69D389BB7D57BE09C861CDBD5CB22E0735980140FBDD564CD
841FD5A012C959261F913F5819557FD7A61D31D007F72FB6D478E1E46E3294C54DFE008FF5F8BE67B6051BB19EDED
7C98B5A8681218FB5F6FD5A6A3E0096967DD8AA002AB8C51E1
REM
tlssse -c -H #?31?key% -H #c20 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-
SHA256 -H psk%PSK% -H identity%id%
tlssse -c -H #R2000P% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlssse -c -H #R2001Q% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlssse -c -H #R2002DP1% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlssse -c -H #R2003DQ1% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
```

```

tlsse -c -H #R2004%PQ% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #P2000%MOD% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #P2001%EXP% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #s20%SIGN% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #?33?type%2000%keyid% -H #?33?type%2001%keyid% -H #?33?type%2080%keyid% -H
#?33?type%2081%keyid% -H #?33?type%2082%keyid% -H #?33?type%2083%keyid% -H
#?33?type%2084%keyid% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
tlsse -c -H #c20 -H #?34%blob0% -H #?34%blob1% -H #?34%blob2% -H #?34%blob3% -H #?34%blob4% -
H #?34%blob5% -H #?34%blob6% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-
SHA256 -H psk%PSK% -H identity%id%
REM
tlsse -c -H #s20%V32% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H
psk%PSK% -H identity%id%
echo Refx: %REF%
REM
PAUSE

```

```

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 908 ms
ClientSession with key1.com opened in 908 ms

```

```

Send: ?311234567890ABCDEF1234567890ABCDEF
Recv: OK
in 328 ms
Send: c20
Recv: OK
in 78 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 817 ms
ClientSession with key1.com opened in 817 ms

```

```

Send:
R2000A6DBBA1977E559876A3F42A1AFFD57AA1F402494E22BA17523E4F2729D3145DFBCDE29431A774C30DE6D09CD8
DF8299F1355CFEEC9A3824711C1F3E4B3FFCE02668DEF4CA5085873E10C59C2E68074C208A04297CE081F56679E092
31970D303A56238F7ABF0CB2E09B8BF9A51346EEE3D90A3E2346B83A1D3ED00FFCD36AF6F
Recv: OK
in 599 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 811 ms
ClientSession with key1.com opened in 811 ms

```

```

Send:
R2001FE9644F0A6D0EF464238F38B53AF907474C5B12FE637E0095851F17AB1FEA9BD9C745AE43BAEB0664969C2E5B
09121DC07D72235679CE7B36BB50AF74B2258724CE5D4E15F7EAE9D7A21F398EF1634606504B842A6076D6E0CD9B04
A3E0116063EBC3F949324849B2409CD428D2D68959CA4C231D76FC16BB6AAA7EB35DEA9B7
Recv: OK
in 590 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 836 ms
ClientSession with key1.com opened in 836 ms

```

```

Send:
R20028288C6F001B1F4F651122848616BAA5736977186DFE10E309BE21221C403B60C30AF044CB0B9BF6B43B6BDC28
75C2B037D96D65292804DFCD7518A41D2E73DFCF6C9F6F958EA4C786B08C4780AE757702C5B637D6AFB43B8CEDAFD7
EEDFA2BBA607994E37D29A569E0C51D6FD63354CD2070B524D3A98092EB3E3057EDECE815
Recv: OK
in 597 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module

```

TLS1.3 Client key1.com is connected in 837 ms
ClientSession with key1.com opened in 837 ms

Send:
R20031AD2C856B6ACC2D91B3716E9F21A48550CCC9B7B997CFF9CDC725DC7F8A3054D5603A863EDA9DD2A19C80BE1D
975A124A6AAD30EDF68D7BCBEB7677D60AC30B77244B0E642027DF157A1E62191E73C0380013C4CC1637CF320657B3
E8743E4072A6AF923F550DEABDDE32E7E3993B4363737E0C21199D3BE3E93C27774C88E07

Recv: OK
in 598 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 816 ms
ClientSession with key1.com opened in 816 ms

Send:
R20047EE0068F1F40563006B86799780B5B3E0ADF9243DA2A51FE854CF8032C5E1DE45EE290757E68D7F5FBFEDD53E
7B357C435DA908D3C5250F5842BD3178DE39C063D1B0E01B094537FA5A5E8AB8E0ABB8E8E996A517690DCE42F0E0C5
245AD052E880B6BCEAE6B6E9DB4C387280D2C7EAA826F1BA91CC5370D46E138E5C91860D5

Recv: OK
in 584 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 834 ms
ClientSession with key1.com opened in 834 ms

Send:
P2000A5EFF4538809B418C9FB7D6EFC8E6528CFA3DAF500B992D7F8CED705933D3E230590DD54C2E2A5BCE4C1F0F48
F716D95B43A0E9AFB6D4F3E1CF4CB8F2C7D9D10E27104A89817419296E93D6FF1D6B9900CC8D7C82D5653C446762C6
BFC4B470AF36D6B997C9D272D0F62AB698997F4433B5226C6E804B13202108ECFAF8CB9253B35C96213C75B54E696DA
2E8A08610698D3D4A702DEE6DE5BA450DD82BD5120C3373285EF5F0CFF7CEFA571167EFFDE517EB8F48E1A477582A
1F3DCE5A6BDE2E5FFF7569BB8B29223ADE10BF6D0D1C4B1AECB42EDB5D9B4FE5C02F1CB7E2F3A346D11E29D6DB48D3
CBEF621F83525296EE3B29D102FB9703EBF50F7E929AF59

Recv: OK
in 879 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 849 ms
ClientSession with key1.com opened in 850 ms

Send: P2001010001
Recv: OK
in 289 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 827 ms
ClientSession with key1.com opened in 827 ms

Send: s201234567890ABCDEF1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF
Recv:
403B383E773512176786574CE741AD51133E9AC5866E7C7FE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD1E04
ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C
3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DED1F1
241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D4531729
75101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC54A512CFD15810E6B
AF17525865F3D0D8771522235F887C7BD7503C5CB8

in 903 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 806 ms
ClientSession with key1.com opened in 806 ms

Send: ?33002000E1E2E3E4
Recv:
002000E1E2E3E47FA49FFEB9AC2A0C130FA3B4FBEF90471265123E0243C63BD7D434B0AB9865A1506AFD252CC894BF
50646DEF5555356563682AFCCA1C0C5599FB0C220BD30AF9C146DE501EA5BE646CB3A99C68638593490EFFF8400F3B
929C61B90565009614272673A35AEC44AA02426238ABEBE806A6E74D18F93D41F4DEE57C65AD9EC4C89F5ACD6AB589
4E7C0BC6F5CF4D99CE6AB204564462A9897FAF74DB069B882A08C1297E1408A56EDCC133FF17E4124E7FA729C7DF9D

319CC96F169DEFBD33F905C60D8033E5DBD7895251DA340602553512C3686C82CBE7AC25671475041FAF00B1FE04B5
2FA4A5C94A9DE169A67C8FE43137C22B1CF94672404A26ACECCAF0D12BBC9196F73DFC219CE9043C2FD05A81F2846E
58AA35E5596189FF0F
in 1002 ms
Send: ?33002001E1E2E3E4
Recv: 002001E1E2E3E42BFA610604774B8A8ED4D4523601E25436BD50BFBB2AEBA64A43072443AFE
in 188 ms
Send: ?33002080E1E2E3E4
Recv:
002080E1E2E3E4649F4618DBFEAE9B7DEF8B052D720F50F84E30FC1564590F250049CC2387F15815100286144516EF
9173C13E8EC94E8742ADC143A3BC919AEE64C58AA99B03543C1F98A10086ADD7754906C74A208E50913ED1B428DF5E
097BC845D20CBB56297DE3CEF4E552239093A42D7481EC66B3F329AFDF11053FED07C5658AB7BA40B58AB85A732ED0
7F8E9669BB7510C1CAA6E319FDA2DD5BF3E090893902
in 503 ms
Send: ?33002081E1E2E3E4
Recv:
002081E1E2E3E4E194395D570D0BD5E6EB0A87A438AD86A7723A28ECE7AFF209050A9F8E494EE6E3361DED1F3EAEF3
1529DDCA4A3AC6FF0D94727E682A3CB5DEC222A1BF625E9A0057421F3E081E50D384A16B434CB41E73EA02331BF637
6178A1F6F871E12BA044D0F6CA40EAF0FDB5A5F39A40CA45A78D7EF68C675E5C23F47EE1791F80F2ED43E723336FA2
280418D4FAB24B22DD2BE2FE88F363C5606EDA66EC00
in 486 ms
Send: ?33002082E1E2E3E4
Recv:
002082E1E2E3E456C85493AD66729A11D20D7C4F69205454DCF8D69F09F3663A94DCC4B7F41F35062EDCD7FA64DDEC
1022C588B06DDC375D58B14B1269238662AF5A8CE120E31CFE9D35EA2D6DFBE8C01961CACF738AAFF2B01D8A3566BA
F61978C3BD0F0402E47344CBA692A35C496C010F7D7F131731746D7AAF99F3552563283CBB7FC8AF8FBA7F3A2FCEA1
0B134444F6BBD13EE1D3CA8A445A947A5FCBF066B50C
in 488 ms
Send: ?33002083E1E2E3E4
Recv:
002083E1E2E3E438489E06DD4A492A29C411EE13FC1815D7C94D008A4CEFAF1BCBB988AA2B6FE23BC2A00565200E21
39BB962B34F90920DDAD0DDCF5F714504741E9176D815BB4A9ABA2DFC5751D8DC422D1B9D673170772C5AC27D37743
D3888F885A917DDFEAD96E2FB444C1C7A93B81D639FCF0869DEB851D576AE66C0C5168B2EF2BA3AB5A2AD6FCA63939
6232E03697E1857AB4DE94A7A59715A34450656C1A65
in 500 ms
Send: ?33002084E1E2E3E4
Recv:
002084E1E2E3E4EFCFFB15347E0BDA2738E95515767D92AA6B341313F4BF300143CDF33EFC3E40EE2EFAE468BE81A5
E8E524899480BF5A223A9AD97BCBFD55DDFF8BA70728A4BAE4C373D8211FB7B05C48B91E85132041AD144BE80AFF4
1D871EE5291B370462FDF4D999BFE95E601A9419E09D95BF773C30827A69724968B4297F601C99B30E4D2D41AF354
4A5B380409A6157B35BCB5E46DFA2DA532185ECA121
in 487 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 806 ms
ClientSession with key1.com opened in 806 ms

Send: c20
Recv: OK
in 287 ms
Send:
?34002000E1E2E3E4339A818162E9790A1D9BEE70C6A1838D037BD8A4E479653E86A566E9A3BA90462DA19206F6557
2D48BF546396B5C56849596DE63901517C2969FD557C96ABF127CCC304A1C40B19964FA219645087CBCBD7CD32E816
864CAA04C36BE24C31E0DD751F084E21F3E9182FC6E42705009BB9017623CC9DE4507FC4D5FAEBC99FC7B4041F8CF4
C98A11D8F9C680752D9582A86C4EFED0100C9A0FCEE1A0C9DA6A15477311C6A7E97D45B5D447231805C50F29125775
9737EF13C91DD2D3210D7880FDAC02255CC661648D1B3998DC45E951328F64D24116A51B1B40A78C422D8A66641FBF
714133B649959CEA8274C5199145836679CD581AF51A014A64C7165D252627699CC94E39E87BF0A022A48FEC2A7A7E
6440BB6B52748BB0A7FFA
Recv: OK
in 897 ms
Send: ?34002001E1E2E3E4295BAF58E4F23CF9525A79792E21E8BC123EE87EB3A56CB692850B782B2DEE
Recv: OK
in 204 ms
Send:
?34002080E1E2E3E4FD836F4961770D56B04ED86D4C2FBE41B62880613DC84ECB86ACAFFFE5397EBB9D543ECC4F4F6
DF9A5E12C8F1DDF0BDA45E97A460354D38FB81F04A48E5B2199A52E2FAE33BB9F048541CD8C8811EA91ED584996015
59B9E263529534AB87C2DB3A6FD622D4ADE5B968FFA80B132C6F91AD612C5E39E7F082B524FB94E001B260BE0F92B8
03EF00505D78D67D44183A9568BA4FDBFEBFC1CC01778E2
Recv: OK
in 553 ms
Send:
?34002081E1E2E3E4A7D12543C97895112E1EAA8A7E1EB9BABF56DE9CCA270D1A83472C54A2C52FEEE6F07604921AE
095009458561D8F064BF9990082777AEDC53B105A8E6C75D1D44F7CB725975E8441ABDBEC08BE12298A340CD2D49C
C370E88AC964074B934C52BEB1303FB6E75515D0ECD18731F266EFD4DE2A5E2AA7EF983FF94621C77E8AC7530157A
B71ABC1E00703EFB9B6255BAC9D2A09CB53726580824D8A

```

Recv: OK
in 540 ms
Send:
?34002082E1E2E3E4E00BE9D6A32F9715D065E9629527F8C6366D494CCB60DD1AC3B666EA9A94CCBA177D9BC7F28E2
D2DC6E71996F09930197E83C92FC1CE4EA34E1EB2928CFD851CF9324C0A62B541F8A68ED9407CE0B340E1E037C7E41
DB4ECFFBF666DA94CC78776D9138C5DBDECCD5A1783E9D80FB1D49823617369B9C05860D78ADB421FB4FFBB23EDF91
789D3A05C6C547301722B063D1B4CAB5B82F70E6AB3DA6F
Recv: OK
in 547 ms
Send:
?34002083E1E2E3E4C18D255E9F2C0F4AF9F94D978BA318AFF40C6FFCE0A0EF810FBE40CFC2AF14C6081953C7DFEE1
A1F0099FA16E6D5F47F61BE254C729549D49BA15422C2DF6F0CB3D9E9612380ED4586DCAFF9F92862A40FAF0416220
F9028838960B4F767240E8B932CBF4BAC7D7E0783DDAA7F71D2CC8ACADBC0B49BAB17DAC8219438903750F6024B84
2BEE47968D189577AD5498698F2D2EC157A691D5E6FB025
Recv: OK
in 533 ms
Send:
?34002084E1E2E3E4CA9BF77B6F8CCB7A55EF84B408DB6155831DEFF31F36746BD9BBF5CABC1B3354941A425D36128
10D27BD63940530E5E3E0D373C9D2AB08D0E2B248B69D389BB7D57BE09C861CDBD5CB22E0735980140FBD564CD841
FD5A012C959261F913F5819557FD7A61D31D007F72FB6D4D78E1E46E3294C54DFE008FF5F8BE67B6051BB19EDEB7C9
8B5A8681218FB5F6FD5A6A3E0096967DD8AA002AB8C51E1
Recv: OK
in 549 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 801 ms
ClientSession with key1.com opened in 801 ms

Send: s201234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef
Recv:
403B383E773512176786574CE741AD51133E9AC5866E7C7FE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD1E04
ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C
3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DED1F1
241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D4531729
75101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC54A512CFD15810E6B
AF17525865F3D0D8771522235F887C7BD7503C5CB8
in 909 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
Refx:
403B383E773512176786574CE741AD51133E9AC5866E7C7FE7A3DBE47453E0095DC0C505DC004FA8445B2AA96DD1E04
ED1DED4CC681B13BB1D7A2A8A246EA1B63598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C
3F0490A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001B961D2C706B7F3710863DED1F1
241399C08C09209B760AA596605BD9446040BF3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D4531729
75101270EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6182FC54A512CFD15810E6B
AF17525865F3D0D8771522235F887C7BD7503C5CB8

```

2.19 test_export_bip32

```

@echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
set type=00
set keyid=E1E2E3E4
REM
set key=1234567890ABCDEF1234567890ABCDEF
REM
set
seed=f0acfacc9279b18275f6923a1f2d2c2be622e39e7bda9ed6d8a783f200b4a011793df8cde50014b5d87169712
21dc7c63e6ff4e22128efa314b4c32fc07e76ae
REM path m/0'/1'/2'/3'
set path=80000000800000018000000280000003
REM
set
blob=003800E1E2E3E4332ACF335377D1617AF0A8BEB1D43F479C56221410D36FF410A8A213D31091B98E1091C5FBB
4A7F02A4EE36334CE1291DF2E1C1C010DFCD303161CAE0A2A30ACF5346F0AE15673A7A9F0E9FC843B584B7FB9882FA
4109CBDA93DB191
REM

```

```
tlssse -c -H #?31?key% -H #c38 -H #t38%seed% -H #v38 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l
TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
tlssse -c -H #?32?type%3800%keyid% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
tlssse -c -H #?33?type%3800%keyid% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
tlssse -c -H #c38 -H #v38 -H #?34%blob% -H #v38 -H #b00%path% -H #p00 -H #?02 -S %SEN% -s -p
%PORT% -h %IP% -l TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
REM
PAUSE
```

```
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 902 ms
ClientSession with key1.com opened in 902 ms
```

```
Send: ?311234567890ABCDEF1234567890ABCDEF
Recv: OK
in 318 ms
Send: c38
Recv: OK
in 78 ms
Send:
t38f0acf9279b18275f6923a1f2d2c2be622e39e7bda9ed6d8a783f200b4a011793df8cde50014b5d8716971221
dc7c63e6ff4e22128efa314b4c32fc07e76ae
Recv: OK
in 242 ms
Send: v38
Recv:
F0ACFACF9279B18275F6923A1F2D2C2BE622E39E7BDA9ED6D8A783F200B4A011793DF8CDE50014B5D8716971221DC7
C63E6FF4E22128EFA314B4C32FC07E76AE
in 174 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 803 ms
ClientSession with key1.com opened in 803 ms
```

```
Send: ?32003800E1E2E3E4
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 794 ms
ClientSession with key1.com opened in 794 ms
```

```
Send: ?33003800E1E2E3E4
Recv:
003800E1E2E3E44C47684333808D6C2D4402932AE5D4250D88AB41E578C093E4342E2C96F87397848E2444246F7729
053D579051DF7760D5573B6C3AF959979128A90877081944EB4F13833ACE461B1A1356338FCBCEE8FFDD54CA4D9AD8
9EB5265ADE
in 524 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 825 ms
ClientSession with key1.com opened in 825 ms
```

```
Send: c38
Recv: OK
in 267 ms
Send: v38
Recv: ERROR
in 65 ms
Send:
?34003800E1E2E3E4332ACF335377D1617AF0A8BEB1D43F479C56221410D36FF410A8A213D31091B98E1091C5FBB4A
7F02A4EE36334CE1291DF2E1C1C010DFCD303161CAE0A2A30ACF5346F0AE15673A7A9F0E9FC843B584B7FB9882FA41
09CBDA93DB191
Recv: OK
in 382 ms
Send: v38
Recv:
F0ACFACF9279B18275F6923A1F2D2C2BE622E39E7BDA9ED6D8A783F200B4A011793DF8CDE50014B5D8716971221DC7
C63E6FF4E22128EFA314B4C32FC07E76AE
in 175 ms
```

```

Send: b0080000000800000018000000280000003
Recv: OK
in 328 ms
Send: p00
Recv:
0482E66E0390200BF7264F7A3C02ED24767BAEEDA9D9205C171577AAF155D3CB32C28EA997BD8905F8F7446269C88A
849EDB5CB83D3B17009560FCAFA8199C466E
in 188 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com

```

2.20 test_export_secret

```

@echo off
set SEN=key1.com
set IP=127.0.0.1
set PORT=8888
REM
set PSK=0102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
set id=Client_identity
REM
set key=1234567890ABCDEF1234567890ABCDEF
set type=00
set keyid=E1E2E3E4
REM
set seed=1234567890ABCDEF1234567890ABCDEF
REM m/0'/1'/2'/3'
set path=800000000800000018000000280000003
REM
set
blob=003000E1E2E3E45EAE2CC17EFE856DBC8FAE94AB157258E1632ADBBBD3E4A2A7D95FF14A9CF38C4FD054070FC
46FC4C6C205A5
REM
tlsse -c -H #?30?key% -H #c30 -H #t30?seed% -H #v30 -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l
TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #?32?type%3000?keyid% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #?33?type%3000?keyid% -H #?02 -S %SEN% -s -p %PORT% -h %IP% -l TLS13-AES128-
CCM-SHA256 -H psk%PSK% -H identity%id%
tlsse -c -H #c30 -H #v30 -H #?34?blob% -H #v30 -H #b00?path% -H #?02 -S %SEN% -s -p %PORT% -h
%IP% -l TLS13-AES128-CCM-SHA256 -H psk%PSK% -H identity%id%
REM
PAUSE

```

```

ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 884 ms
ClientSession with key1.com opened in 885 ms

```

```

Send: ?301234567890ABCDEF1234567890ABCDEF
Recv: OK
in 325 ms
Send: c30
Recv: OK
in 70 ms
Send: t301234567890ABCDEF1234567890ABCDEF
Recv: OK
in 110 ms
Send: v30
Recv: 1234567890ABCDEF1234567890ABCDEF
in 90 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 813 ms
ClientSession with key1.com opened in 813 ms

```

```

Send: ?32003000E1E2E3E4
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 805 ms

```

ClientSession with key1.com opened in 805 ms

```
Send: ?33003000E1E2E3E4
Recv:
003000E1E2E3E4EEB94BCFFCA0871B3DEEF21BC50EA6B7310192B6A740EFB3B9E4DC767EBD64190783F8F08D576315
3F5C2F88
in 426 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
ConnectTo: key1.com@127.0.0.1:8888
Using Software Identity Module
TLS1.3 Client key1.com is connected in 802 ms
ClientSession with key1.com opened in 802 ms

Send: c30
Recv: OK
in 278 ms
Send: v30
Recv: ERROR
in 62 ms
Send:
?34003000E1E2E3E45EAE2CC17EFE856DBC8FAE94AB157258E1632ADBBBD3E4A2A7D95FF14A9CF38C4FD054070FC46
FC4C6C205A5
Recv: OK
in 232 ms
Send: v30
Recv: 1234567890ABCDEF1234567890ABCDEF
in 97 ms
Send: b00800000008000000180000002800000003
Recv: ERROR
in 101 ms
Send: ?02
Recv: TLS1.3 Disconnected from key1.com
```

3 Test APDU

These tests are only available for TLS-SE test version.

3.1 test_apdu_aesccm

```
/T=0
// select
00 A4 04 00 06 01 02 03 04 05 00
00 20 00 01 08 30 30 30 30 30 30 30
//
00 85 05 05 24 01020304050607080910111213141516 B1B2B3B4B5B6B7B8B9BABBBC A1A2A3A4A5 D1D2D3
// A1A2A3A4A5 55EF87 3FAC1CC4E1042F01CA0B48B72BC88D7E 9000
00 85 05 06 34 01020304050607080910111213141516 B1B2B3B4B5B6B7B8B9BABBBC A1A2A3A4A5 55EF87
3FAC1CC4E1042F01CA0B48B72BC88D7E
//
//key: 000102030405060708090A0B0C0D0E0F
//Nonce: 101112131415161718191A1B
//Message: 202122232425262728292A2B2C2D2E2F30313233
//Cipher: 03949B8366D10B95228555EB5849F5A94B86E084
//Length: 20 bytes
//
//INS=85 P1=AD_Length P1=00=>05 P1=FF=>noAD P2=05=Encrypt P3=Length AES_Key IV AD Data
//INS=85 P1=AD_Length P1=00=>05 P1=FF=>noAD P2=06=Decrypt P3=Length AES_Key IV AD Data Tag
//
00 85 0E 05 3E 000102030405060708090A0B0C0D0E0F 101112131415161718191A1B
A0A1A2A3A4A5A6A7A8A9AAABACAD 202122232425262728292A2B2C2D2E2F30313233
//A0A1A2A3A4A5A6A7A8A9AAABACAD 03949B8366D10B95228555EB5849F5A94B86E084
A269DC2E14F5067C2E4514BB0ED4D129 9000
00 85 0F 05 3F 000102030405060708090A0B0C0D0E0F 101112131415161718191A1B
A0A1A2A3A4A5A6A7A8A9AAABACADAE 202122232425262728292A2B2C2D2E2F30313233
//A0A1A2A3A4A5A6A7A8A9AAABACADAE 03949B8366D10B95228555EB5849F5A94B86E084
B98ECFDAFE5FD28EECEE057A593FE0AC 9000
00 85 10 05 40 000102030405060708090A0B0C0D0E0F 101112131415161718191A1B
A0A1A2A3A4A5A6A7A8A9AAABACADAEAF 202122232425262728292A2B2C2D2E2F30313233
```

```
//A0A1A2A3A4A5A6A7A8A9AAABACADAEAF 03949B8366D10B95228555EB5849F5A94B86E084
A011ACEDECA122123D0F449F9FEF4698 9000
00 85 1E 05 4E 000102030405060708090A0B0C0D0E0F 101112131415161718191A1B
A0A1A2A3A4A5A6A7A8A9AAABACADAEAFB0B1B2B3B4B5B6B7B8B9BABBBCBD
202122232425262728292A2B2C2D2E2F30313233
//A0A1A2A3A4A5A6A7A8A9AAABACADAEAFB0B1B2B3B4B5B6B7B8B9BABBBCBD
03949B8366D10B95228555EB5849F5A94B86E084 2C7A47C3AC075B8A3E041E4BF010A99D 9000
00 85 1E 06 5E 000102030405060708090A0B0C0D0E0F 101112131415161718191A1B
A0A1A2A3A4A5A6A7A8A9AAABACADAEAFB0B1B2B3B4B5B6B7B8B9BABBBCBD
03949B8366D10B95228555EB5849F5A94B86E084 2C7A47C3AC075B8A3E041E4BF010A99D
//202122232425262728292A2B2C2D2E2F30313233 9000
00 85 FF 05 30 000102030405060708090A0B0C0D0E0F 101112131415161718191A1B
202122232425262728292A2B2C2D2E2F30313233
//03949B8366D10B95228555EB5849F5A94B86E084 E904DBA99D34B6DD630EA50C2BC992CB
00 85 FF 06 40 000102030405060708090A0B0C0D0E0F 101112131415161718191A1B
03949B8366D10B95228555EB5849F5A94B86E084 E904DBA99D34B6DD630EA50C2BC992CB
//202122232425262728292A2B2C2D2E2F30313233 9000
00 85 FF 05 1C 000102030405060708090A0B0C0D0E0F 101112131415161718191A00
//8E80D99CF29C7004F81179DE65789C56
00 85 FF 06 2C 000102030405060708090A0B0C0D0E0F 101112131415161718191A00
8E80D99CF29C7004F81179DE65789C56
//9000
/Q
```

SuperReader 1.5
 Copyright (C) 2003,2025 Pascal Urien (Pascal.Urien@gmail.com)
 All Rights Reserved.
 List of Active Readers
 #0,127.0.0.1:7777/0

127.0.0.1:7777/0 is used
 type exit to leave

```
Reader>Smartcard Inserted...Protocol T=1
ATR: 3B 07 52 41 43 53 30 30 31
Reader>
Loading script scriptdefault.txt ...OK
```

```
> 00 A4 04 00 06 01 02 03 04 05 00
< 90 00
> 00 20 00 01 08 30 30 30 30 30 30 30 30
< 90 00
> 00 85 05 05 24 01 02 03 04 05 06 07 08 09 10 11
12 13 14 15 16 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB
BC A1 A2 A3 A4 A5 D1 D2 D3
< A1 A2 A3 A4 A5 55 EF 87 3F AC 1C C4 E1 04 2F 01
CA 0B 48 B7 2B C8 8D 7E 90 00
> 00 85 05 06 34 01 02 03 04 05 06 07 08 09 10 11
12 13 14 15 16 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB
BC A1 A2 A3 A4 A5 55 EF 87 3F AC 1C C4 E1 04 2F
01 CA 0B 48 B7 2B C8 8D 7E
< A1 A2 A3 A4 A5 D1 D2 D3 90 00
> 00 85 0E 05 3E 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
1B A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD 20
21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30
31 32 33
< A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD 03 94
9B 83 66 D1 0B 95 22 85 55 EB 58 49 F5 A9 4B 86
E0 84 A2 69 DC 2E 14 F5 06 7C 2E 45 14 BB 0E D4
D1 29 90 00
> 00 85 0F 05 3F 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
1B A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE
20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F
30 31 32 33
< A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE 03
94 9B 83 66 D1 0B 95 22 85 55 EB 58 49 F5 A9 4B
86 E0 84 B9 8E CF DA FE 5F D2 8E EC EE 05 7A 59
3F E0 AC 90 00
> 00 85 10 05 40 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
1B A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE
AF 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E
2F 30 31 32 33
```

```

< A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE AF
03 94 9B 83 66 D1 0B 95 22 85 55 EB 58 49 F5 A9
4B 86 E0 84 A0 11 AC ED EC A1 22 12 3D 0F 44 9F
9F EF 46 98 90 00
> 00 85 1E 05 4E 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
1B A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE
AF B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB BC BD 20
21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30
31 32 33
< A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE AF
B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB BC BD 03 94
9B 83 66 D1 0B 95 22 85 55 EB 58 49 F5 A9 4B 86
E0 84 2C 7A 47 C3 AC 07 5B 8A 3E 04 1E 4B F0 10
A9 9D 90 00
> 00 85 1E 06 5E 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
1B A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE
AF B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB BC BD 03
94 9B 83 66 D1 0B 95 22 85 55 EB 58 49 F5 A9 4B
86 E0 84 2C 7A 47 C3 AC 07 5B 8A 3E 04 1E 4B F0
10 A9 9D
< A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE AF
B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB BC BD 20 21
22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31
32 33 90 00
> 00 85 FF 05 30 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
1B 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E
2F 30 31 32 33
< 03 94 9B 83 66 D1 0B 95 22 85 55 EB 58 49 F5 A9
4B 86 E0 84 E9 04 DB A9 9D 34 B6 DD 63 0E A5 0C
2B C9 92 CB 90 00
> 00 85 FF 06 40 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
1B 03 94 9B 83 66 D1 0B 95 22 85 55 EB 58 49 F5
A9 4B 86 E0 84 E9 04 DB A9 9D 34 B6 DD 63 0E A5
0C 2B C9 92 CB
< 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F
30 31 32 33 90 00
> 00 85 FF 05 1C 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
00
< 8E 80 D9 9C F2 9C 70 04 F8 11 79 DE 65 78 9C 56
90 00
> 00 85 FF 06 2C 00 01 02 03 04 05 06 07 08 09 0A
0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A
00 8E 80 D9 9C F2 9C 70 04 F8 11 79 DE 65 78 9C
56
< 90 00

```

End of script

3.2 test_apdu_ecc

```

/T=0
// select
00 A4 04 00 06 01 02 03 04 05 00
//AdmPIN
0020 0001 08 30 30 30 30 30 30 30 30

// P2=Key index
// Clear Key2
00 81 00 02 00
// init curve P1=1= SECP256v1 index=P2=01
// P1=0xFF for SECP256r1
00 89 01 02 00
// Set Private Key
00 88 07 02 20 CFEBA5FB779C84ED89EF364B892E916F52CE6BC20F3A856129EEDE4D1D07BDCB
// SET Pub Key
00 88 06 02 41 04
37232040741008CF078D96BC8EAF76365FC6A98AF30203B6022739813677FA26C1E014A5F8CFA67DD0DB7F97B9120
23B86063E3B9BEADC95D9F1C9CCD1209B6
// Get Public Key

```

```

00 84 06 02 00
// Get Private Key
00 84 07 02 00
// ECDSA RAW, P1=00
00 80 00 02 20 0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF

// Clear Key1
00 81 00 01 00
// init curve SECP256k1
00 89 FF 01 00
// Genkey P2=index
00 82 00 01 00
// Get Public Key
00 84 06 01 00
// Get Private Key
00 84 07 01 00
// ECDSA SHA256, P1=21
00 80 21 01 02 1234
//
Loading script test_apdu_ecc.txt ...
// select
Sending
00A40400006010203040500
Received
9000
//AdmPIN
Sending
0020000108303030303030303030
Received
9000
// P2=Key index
// Clear Key2
Sending
0081000200
Received
9000
// init curve P1=1= SECP256v1 index=P2=01
// P1=0xFF for SECP256r1
Sending
0089010200
Received
9000
// Set Private Key
Sending
0088070220CFEBA5FB779C84ED89EF364B892E916F52CE6BC20F3A856129EEDE
4D1D07BDCB
Received
9000
// SET Pub Key
Sending
00880602410437232040741008CF078D96BC8EAF36365FC6A98AF30203B6022
739813677FA26C1E014A5F8CFA67DD0DB7F97B912023B86063E3B9BEADC95D9F
1C9CCD1209B6
Received
9000
// Get Public Key
Sending
0084060200
Received
00410437232040741008CF078D96BC8EAF36365FC6A98AF30203B6022739813
677FA26C1E014A5F8CFA67DD0DB7F97B912023B86063E3B9BEADC95D9F1C9CCD
1209B69000
// Get Private Key
Sending
0084070200
Received
0020CFEBA5FB779C84ED89EF364B892E916F52CE6BC20F3A856129EEDE4D1D07
BDCB9000
// ECDSA RAW, P1=00
Sending
00800002200123456789ABCDEF0123456789ABCDEF0123456789ABCDEF012345
6789ABCDEF
Received
0046304402201550DD3A406D5C75E3271F704724018F3513A6F38993F3FE54AF
494AA2C028780220758BC0DF07D19950DD9D1C6AE06936B768B2D6D0188BDEF4
D98C6041899E85279000
// Clear Key1

```

```

Sending
0081000100
Received
9000
// init curve SECP256k1
Sending
0089FF0100
Received
9000
// Genkey P2=index
Sending
0082000100
Received
9000
// Get Public Key
Sending
0084060100
Received
0041040239499CF107BC8924A1C5E38A2AE72FB8AF0C9018764B6B121E864398
3BEB3B9CBA5EF44B66120511210B2B7875FBDD68F3A1D19A940DD70F7830D09E
F8FCFE9000
// Get Private Key
Sending
0084070100
Received
002046F31EAA86449FD6F749DD3C936CA6D1278B09E2FE5A4BF8D4F31C200862
75BD9000
// ECDSA SHA256, P1=21
Sending
00802101021234
Received
004730450220636AE1CC59403E53A9D01D89AC7356DD421AEC6A377E214499BA
2D82C6D5985B0221008CA3BEE45F6C7FBB471D9FE519138BAA870F316A90A508
9F5F189438798E5AE59000

```

Time: 797 ms

3.3 test_apdu_hmac

```

/T=0
// select
00 A4 04 00 06 01 02 03 04 05 00
//AdmPIN
0020 0001 08 30 30 30 30 30 30 30 30

// Status
00 87 00 00 04

// SHA256
//HMAC-SHA-256 = b0344c61d8db38535ca8afceaf0bf12b
// 881dc200c9833da726e9376c2e32cff7
00 85 00 02 1E 14 0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b 08 4869205468657265

// A.1 RFC 5869 Test Case 1
// =====
// PRK = HMAC(salt, IKM)
// IKM = 0x0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b (22 octets)
// salt = 0x000102030405060708090a0b0c (13 octets)
// PRK = 077709362c2e32df0ddc3f0dc47bba6390b6c73bb50f9c3122ec844ad7c2b3e5
00 85 00 02 25 0D 000102030405060708090a0b0c 16 0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b0b

// T(1) = HMAC-Hash(PRK, T(0) | info | 0x01)
// info= 0xf0f1f2f3f4f5f6f7f8f9 (10 octets)
// L=42
00 85 00 02 2D 20 077709362C2E32DF0DDC3F0DC47BBA6390B6C73BB50F9C3122EC844AD7C2B3E5 0B
f0f1f2f3f4f5f6f7f8f9 01
// 3CB25F25FAACD57A90434F64D0362F2A2D2D0A90CF1A5A4C5DB02D56ECC4C5BF
// T(1) = HMAC-Hash(PRK, T(0) | info | 0x02)
00 85 00 02 4D 20 077709362C2E32DF0DDC3F0DC47BBA6390B6C73BB50F9C3122EC844AD7C2B3E5 2B
3CB25F25FAACD57A90434F64D0362F2A2D2D0A90CF1A5A4C5DB02D56ECC4C5BF f0f1f2f3f4f5f6f7f8f9 02
// 34007208d5b887185865

// rfc 8448
// {server} extract secret "early"

```



```

// Derive-Secret_Early(ESK,"c e traffic", 0s)
0085 0002 38 20 23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897 16 0020
11746c733133206320652074726166666963 00 01
// Derive-Secret(ESK,"c e traffic", 0s)
0085 000B 03 0020 00
//0738A2B6F6FAA2AF5CDD9B6F0F2B232F19B3256A5926EAC600B911F91E98D2D4

// Derive-Secret(ESK,"e exp master", 0s)
0085 0002 39 20 23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897 17 0020
12746c733133206520657870206d6173746572 00 01
// Derive-Secret(ESK,"e exp master", 0s)
0085 010B 03 0020 00
// 9B7FC6A8F854C16A301DFC566859931DB5EE9A22793142A0C67159C445E7BEAB

// Extract(DSK,DHE) = HMAC(DSK,DHE)
0085 0002 42 20 E8E7AC087158FC8440E41A12989F9194783764CD5FC36564028037F2C8206E96 20
97C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D0C85E2F8BF43AB
// HMAC(DSK,DHE) = Extract(salt=DSK,DHE)
0085 000E 20 97C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D0C85E2F8BF43AB
//98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BD

// ESK= HMAC(0,IKM)
00 85 00 02 23 01 00 20 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17
18 19 1A 1B 1C 1D 1E 1F 20
// 23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897

// eBSK= Derive-Secret_Early(ESK, "ext binder" , "" ) = ""= SHA256(NULL)
// Null sha256 E3B0C44298FC1C149AFBF4C8996FB92427AE41E4649B934CA495991B7852B855
0085 0002 57 20 23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897 35 0020 10
746c733133206578742062696e646572 20
E3B0C44298FC1C149AFBF4C8996FB92427AE41E4649B934CA495991B7852B855 01
//4351F8A53AA85AC394AB04C516464CAB96E9340C269632D09899537887EE651F

// DSK = Derive-Secret(ESK, "derived", "")
0085 0002 54 20 23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897 32 0020 0d
746c7331332064657269766564 20 E3B0C44298FC1C149AFBF4C8996FB92427AE41E4649B934CA495991B7852B855
01
// E8E7AC087158FC8440E41A12989F9194783764CD5FC36564028037F2C8206E96

// FSK "tls13 finished" H(BSK, info)
0085 0002 35 20 4351F8A53AA85AC394AB04C516464CAB96E9340C269632D09899537887EE651F 13 0020 0E
746c7331332066696E6973686564 00 01
// FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A4841793139168A65BFCB0

0085 0002 42 20 FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A4841793139168A65BFCB0 20
C0C27BBFDBAF30C5888732734BCE1B47E3F14FAA4BE578A25BD81BBDE2D2B1FA
//BuildTls13HandshakeHmac
//Key FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A4841793139168A65BFCB0
//Data C0C27BBFDBAF30C5888732734BCE1B47E3F14FAA4BE578A25BD81BBDE2D2B1FA
//Result 811ABCE77E614232741AE2F59E0497E8801D31D7388577FAED73EF63BD1B6791

/ HMAC(feBSK,0s)
0085 000C 20 C0C27BBFDBAF30C5888732734BCE1B47E3F14FAA4BE578A25BD81BBDE2D2B1FA
//811ABCE77E614232741AE2F59E0497E8801D31D7388577FAED73EF63BD1B6791

//Extract
//salt (32) E8E7AC087158FC8440E41A12989F9194783764CD5FC36564028037F2C8206E96
//inkey (32) 97C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D0C85E2F8BF43AB
//noutkey (32) 98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BDE

// Extract(DSK,DHE) = HMAC(DSK,DHE)
0085 0002 42 20 E8E7AC087158FC8440E41A12989F9194783764CD5FC36564028037F2C8206E96 20
97C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D0C85E2F8BF43AB
// HMAC(DSK,DHE) = Extract(salt=DSK,DHE)
0085 000E 20 97C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D0C85E2F8BF43AB
//98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BD

// Derive-Secret_Early(ESK,"c e traffic", 0s)
0085 0002 38 20 23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897 16 0020
11746c733133206320652074726166666963 00 01
// Derive-Secret(ESK,"c e traffic", 0s)
0085 000B 03 0020 00
//0738A2B6F6FAA2AF5CDD9B6F0F2B232F19B3256A5926EAC600B911F91E98D2D4

// Derive-Secret(ESK,"e exp master", 0s)
0085 0002 39 20 23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897 17 0020
12746c733133206520657870206d6173746572 00 01

```

46


```

//0AAF8B11FFDF03F29001076627519265C4BC4C0B633B5B185813439614B0B6CB
// eFSK=feBSK "tls13 finished" H(eBSK, info)
Sending
0085000235204351F8A53AA85AC394AB04C516464CAB96E9340C269632D09899
537887EE651F1300200E746C7331332066696E69736865640001
Received
FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A4841793139168A65BFCB0
9000
// FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A4841793139168A65BFCB0
// rFSK=frBSK "tls13 finished" H(rBSK, info)
Sending
0085000235200AAF8B11FFDF03F29001076627519265C4BC4C0B633B5B185813
439614B0B6CB1300200E746C7331332066696E69736865640001
Received
C8CA78E8837F26B758C46F36C99435033E6FC5DB1DA46BE83A0E3E214283F0E3
9000
// C8CA78E8837F26B758C46F36C99435033E6FC5DB1DA46BE83A0E3E214283F0E3
// DSK = Derive-Secret(ESK, "derived", "")
Sending
00850002342023499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409
DE8CD8B3C8971200200D746C73313320646572697665640001
Received
98EEAA27F7D77499E5FBC63A413CD8C395CAE42D850B65A5AE6A63807368A3F5
9000
//98EEAA27F7D77499E5FBC63A413CD8C395CAE42D850B65A5AE6A63807368A3F5
// HMAC(feBSK,data) = HMAC(feBSK, 0s)
Sending
008500022320FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A484179313
9168A65BFCB00100
Received
3E015D850B89C2470D4C49D4BD8E7C76F2B74175DDD85F393569315DA15480A4
9000
// HMAC(feBSK,0s)
Sending
0085000C0100
Received
3E015D850B89C2470D4C49D4BD8E7C76F2B74175DDD85F393569315DA15480A4
9000
//3E015D850B89C2470D4C49D4BD8E7C76F2B74175DDD85F393569315DA15480A4
// HMAC(frBSK,data) = HMAC(frBSK, 0s)
Sending
008500022320C8CA78E8837F26B758C46F36C99435033E6FC5DB1DA46BE83A0E
3E214283F0E30100
Received
7ED9BD7AE073D78CDA4660B70E5E178028A98CCCB1FB0C227F638857912F105B
9000
// HMAC(frBSK,0s)
Sending
0085000D0100
Received
7ED9BD7AE073D78CDA4660B70E5E178028A98CCCB1FB0C227F638857912F105B
9000
//7ED9BD7AE073D78CDA4660B70E5E178028A98CCCB1FB0C227F638857912F105B
// Derive-Secret_Early(ESK,"c e traffic", 0s)
Sending
00850002382023499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409
DE8CD8B3C89716002011746C7331332063206520747261666669630001
Received
0738A2B6F6FAA2AF5CDD9B6F0F2B232F19B3256A5926EAC600B911F91E98D2D4
9000
// Derive-Secret(ESK,"c e traffic", 0s)
Sending
0085000B03002000
Received
0738A2B6F6FAA2AF5CDD9B6F0F2B232F19B3256A5926EAC600B911F91E98D2D4
9000
//0738A2B6F6FAA2AF5CDD9B6F0F2B232F19B3256A5926EAC600B911F91E98D2D4
// Derive-Secret(ESK,"e exp master", 0s)
Sending
00850002392023499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409
DE8CD8B3C89717002012746C733133206520657870206D61737465720001
Received
9B7FC6A8F854C16A301DFC566859931DB5EE9A22793142A0C67159C445E7BEAB
9000
// Derive-Secret(ESK,"e exp master", 0s)
Sending
0085010B03002000

```

```

Received
9B7FC6A8F854C16A301DFC566859931DB5EE9A22793142A0C67159C445E7BEAB
9000
// 9B7FC6A8F854C16A301DFC566859931DB5EE9A22793142A0C67159C445E7BEAB
// Extract (DSK, DHE) = HMAC (DSK, DHE)
Sending
008500024220E8E7AC087158FC8440E41A12989F9194783764CD5FC365640280
37F2C8206E962097C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D
0C85E2F8BF43AB
Received
98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BDE
9000
// HMAC (DSK, DHE) = Extract (salt=DSK, DHE)
Sending
0085000E2097C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D0C85
E2F8BF43AB
Received
98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BDE
9000
//98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BD
// ESK= HMAC (0, IKM)
Sending
00850002230100200102030405060708090A0B0C0D0E0F101112131415161718
191A1B1C1D1E1F20
Received
23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897
9000
// 23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897
// eBSK= Derive-Secret_Early (ESK, "ext binder", "" ) = "" = SHA256 (NULL)
// Null sha256 E3B0C44298FC1C149AFBF4C8996FB92427AE41E4649B934CA495991B7852B855
Sending
00850002572023499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409
DE8CD8B3C89735002010746C733133206578742062696E64657220E3B0C44298
FC1C149AFBF4C8996FB92427AE41E4649B934CA495991B7852B85501
Received
4351F8A53AA85AC394AB04C516464CAB96E9340C269632D09899537887EE651F
9000
//4351F8A53AA85AC394AB04C516464CAB96E9340C269632D09899537887EE651F
// DSK = Derive-Secret (ESK, "derived", "")
Sending
00850002542023499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409
DE8CD8B3C8973200200D746C733133206465726976656420E3B0C44298FC1C14
9AFBF4C8996FB92427AE41E4649B934CA495991B7852B85501
Received
E8E7AC087158FC8440E41A12989F9194783764CD5FC36564028037F2C8206E96
9000
// E8E7AC087158FC8440E41A12989F9194783764CD5FC36564028037F2C8206E96
// FSK "tls13 finished" H (BSK, info)
Sending
0085000235204351F8A53AA85AC394AB04C516464CAB96E9340C269632D09899
537887EE651F1300200E746C7331332066696E69736865640001
Received
FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A4841793139168A65BFBCB0
9000
// FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A4841793139168A65BFBCB0
Sending
008500024220FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A484179313
9168A65BFBCB020C0C27BBFDBAF30C5888732734BCE1B47E3F14FAA4BE578A25B
D81BBDE2D2B1FA
Received
811ABCE77E614232741AE2F59E0497E8801D31D7388577FAED73EF63BD1B6791
9000
//BuildTls13HandshakeHmac
//Key FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A4841793139168A65BFBCB0
//Data C0C27BBFDBAF30C5888732734BCE1B47E3F14FAA4BE578A25BD81BBDE2D2B1FA
//Result 811ABCE77E614232741AE2F59E0497E8801D31D7388577FAED73EF63BD1B6791
Sending
0085000C20C0C27BBFDBAF30C5888732734BCE1B47E3F14FAA4BE578A25BD81B
BDE2D2B1FA
Received
811ABCE77E614232741AE2F59E0497E8801D31D7388577FAED73EF63BD1B6791
9000
//811ABCE77E614232741AE2F59E0497E8801D31D7388577FAED73EF63BD1B6791
//Extract
//salt (32) E8E7AC087158FC8440E41A12989F9194783764CD5FC36564028037F2C8206E96
//inkey (32) 97C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D0C85E2F8BF43AB
//noutkey (32) 98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BDE

```

```

// Extract(DSK,DHE) = HMAC(DSK,DHE)
Sending
008500024220E8E7AC087158FC8440E41A12989F9194783764CD5FC365640280
37F2C8206E962097C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D
0C85E2F8BF43AB
Received
98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BDE
9000
// HMAC(DSK,DHE) = Extract(salt=DSK,DHE)
Sending
0085000E2097C3CF6C99EA9840D2E35F0AE684C97A2B4B68D5F8196C0C1D0C85
E2F8BF43AB
Received
98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BDE
9000
//98F057EE0CC6C17E27983B9C2019111AE1E440517863C803030877139C047BD
// Derive-Secret_Early(ESK,"c e traffic", 0s)
Sending
00850002382023499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409
DE8CD8B3C89716002011746C7331332063206520747261666669630001
Received
0738A2B6F6FAA2AF5CDD9B6F0F2B232F19B3256A5926EAC600B911F91E98D2D4
9000
// Derive-Secret(ESK,"c e traffic", 0s)
Sending
0085000B03002000
Received
0738A2B6F6FAA2AF5CDD9B6F0F2B232F19B3256A5926EAC600B911F91E98D2D4
9000
//0738A2B6F6FAA2AF5CDD9B6F0F2B232F19B3256A5926EAC600B911F91E98D2D4
// Derive-Secret(ESK,"e exp master", 0s)
Sending
00850002392023499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409
DE8CD8B3C89717002012746C733133206520657870206D61737465720001
Received
9B7FC6A8F854C16A301DFC566859931DB5EE9A22793142A0C67159C445E7BEAB
9000
// Derive-Secret(ESK,"e exp master", 0s)
Sending
0085010B03002000
Received
9B7FC6A8F854C16A301DFC566859931DB5EE9A22793142A0C67159C445E7BEAB
9000
// 9B7FC6A8F854C16A301DFC566859931DB5EE9A22793142A0C67159C445E7BEAB
// Extract(DSK,DHE) = HMAC(DSK,DHE)
Sending
008500022320E8E7AC087158FC8440E41A12989F9194783764CD5FC365640280
37F2C8206E960100
Received
7092C2117D67E6AEB5C5FDF5E6D9C70FBDC69B374E914C26AB08A122483D0E73
9000
// HMAC(DSK,DHE) = Extract(salt=DSK,DHE)
Sending
0085000E0100
Received
7092C2117D67E6AEB5C5FDF5E6D9C70FBDC69B374E914C26AB08A122483D0E73
9000
//7092C2117D67E6AEB5C5FDF5E6D9C70FBDC69B374E914C26AB08A122483D0E73
// HMAC(feBSK,data) = HMAC(feBSK, 0s)
Sending
008500022320FCA24690D17DDE3F727D29D2186A5F83E1AEBD4889A484179313
9168A65BFCB00100
Received
3E015D850B89C2470D4C49D4BD8E7C76F2B74175DDD85F393569315DA15480A4
9000
// HMAC(feBSK,0s)
Sending
0085000C0100
Received
3E015D850B89C2470D4C49D4BD8E7C76F2B74175DDD85F393569315DA15480A4
9000
//3E015D850B89C2470D4C49D4BD8E7C76F2B74175DDD85F393569315DA15480A4
// ESK= Extract(0,IKM)
Sending
0085000A230100200102030405060708090A0B0C0D0E0F101112131415161718
191A1B1C1D1E1F20
Received

```

```

23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897
9000
//23499E7EDF0FBE6BAA137DF0F23BECAEFA722AD19FC262855409DE8CD8B3C897 ES

Time: 2910 ms

```

3.4 test_apdu_cert

```

/T=0
// select
00 A4 04 00 06 01 02 03 04 05 00
//ADM
0020 0001 08 30 30 30 30 30 30 30 30

// clear key3
0081 0003 00
// init curve key=3, P2=FF=>SECP256k1
0089 FF03 00
// set PrivKey
0088 0703 20 5CDEE66CAEDE933ADD0CA894393480A076431FCA3EE6AC9A8E1129CE5F794120
// set PubKey
0088 0603 41
046099836D971593AAA2C1C32B6DB9EF9521041795E21CF1E7511DF3BD358F97DF358B33A875E359CBE236163D6DBA
EDFEC6C9393522C7EBC25A7CC85E1F0A7D67
// Write Cert + Autosign
00 8F FF 03 00
// Read Cert
00 8E 00 00 40
// Read App Public Key
00 84 06 FE 65
// SIGN with APP PubKey P2=FE, P1=00 (RAW) P1=21 SHA256
00 80 21 FE 02 12 34
00 80 00 FE 20 1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF1234567890ABCDEF
//

Loading script test_apdu_cert.txt ...
// select
Sending
00A4040006010203040500
Received
9000
//ADM
Sending
00200001083030303030303030
Received
9000
// clear key3
Sending
0081000300
Received
9000
// init curve key=3, P2=FF=>SECP256k1
Sending
0089FF0300
Received
9000
// set PrivKey
Sending
00880703205CDEE66CAEDE933ADD0CA894393480A076431FCA3EE6AC9A8E1129
CE5F794120
Received
9000
// set PubKey
Sending
0088060341046099836D971593AAA2C1C32B6DB9EF9521041795E21CF1E7511D
F3BD358F97DF358B33A875E359CBE236163D6DBAEDFEC6C9393522C7EBC25A7C
C85E1F0A7D67
Received
9000
// Write Cert + Autosign
Sending
008FFF0300
Received
9000

```

```

// Read Cert
Sending
008E000040
Received
A3262715E0CF539CDA43ABF4B3E3DEA17B343445C6B8866C7A90F416D6187257
0258961EB0A045A2514B1A73F3963559315D6404CFAA93F1DF8B028ABFDEE673
9000
// Read App Public Key
Sending
008406FE65
Received
0041045C4D07497E0F6C09812EF9A5E1F53A247742BC028D19EECF445C3C0C2D
5B40D6097A06E2EF102A0E238DF84FC15C265625D43B5058A5F61A378CB670B6
C9296E9000
// SIGN with APP PubKey P2=FE, P1=00 (RAW) P1=21 SHA256
Sending
008021FE021234
Received
0046304402200F60FC9F463AD669B212A0A6AED1881E8ECFD416B47063BD1955
2D60B44262C002207A1FB60C5BED88335CCFFC177E7E21EA8FEF5816BDE739DB
C0C4F7C6329DD6C29000
Sending
008000FE201234567890ABCDEF1234567890ABCDEF1234567890ABCDEF123456
7890ABCDEF
Received
0046304402203BDC6F7F221A1DE6F9C5E7CE69759C1D3EA37C59621A223AAED4
ECF8F2F820D20220128A6703C3ED51C0261E0B17760C1115882FE54539E8C253
2F51B72263303D749000

```

Time: 656 ms

3.5 test_apdu_rsa

```

/T=0
// select
00 A4 04 00 06 01 02 03 04 05 00
// Admun PIN
00 20 00 01 08 30 30 30 30 30 30 30
//STATUS
//332E3030 F70A 1030 08C0 9000
00 87 00 00 00
// CLEAR RSA 0
00 81 00 20 00
// GEN RSA 0
00 82 00 20 00
// get key
00 84 00 20 00
00 84 01 20 00
00 84 02 20 00
00 84 03 20 00
00 84 04 20 00
00 84 05 20 00
00 84 06 20 00
//
// Sign
00 80 00 20 20 1234567890ABCDEF 1234567890ABCDEF 1234567890ABCDEF 1234567890ABCDEF
//
// CLEAR RSA 0
00 81 00 20 00
// GEN RSA 0
00 82 00 20 00
// get key
00 84 00 20 00
00 84 01 20 00
00 84 02 20 00
00 84 03 20 00
00 84 04 20 00
00 84 05 20 00
00 84 06 20 00
//
00 81 00 20 00
// P, Q, DP1, DQ1, PQ

```

```

00 8C 00 20 80
A6DBBA1977E559876A3F42A1AFFD57AA1F402494E22BA17523E4F2729D3145DFBCDE29431A774C30DE6D09CD8DF829
9F1355CFEEC9A3824711C1F3E4B3FFCE02668DEF4CA5085873E10C59C2E68074C208A04297CE081F56679E09231970
D303A56238F7ABF0CB2E09B8BF9A51346EEE3D90A3E2346B83A1D3ED00FFCD36AF6F
00 8C 00 30 80
FE9644F0A6D0EF464238F38B53AF907474C5B12FE637E0095851F17AB1FEA9BD9C745AE43BAEB0664969C2E5B09121
DC07D72235679CE7B36BB50AF74B2258724CE5D4E15F7EAE9D7A21F398EF1634606504B842A6076D6E0CD9B04A3E01
16063EBC3F949324849B2409CD428D2D68959CA4C231D76FC16BB6AAA7EB35DEA9B7
00 8C 00 40 80
8288C6F001B1F4F651122848616BAA5736977186DFE10E309BE21221C403B60C30AF044CB0B9BF6B43B6BDC2875C2B
037D96D65292804DFCD7518A41D2E73DFCF6C9F6F958EA4C7F86B08C4780AE757702C5B637D6AFB43B8CEDAFD7EEDFA
2BBA607994E37D29A569E0C51D6FD63354CD2070B524D3A98092EB3E3057EDECE815
00 8C 00 50 80
1AD2C856B6ACC2D91B3716E9F21A48550CCC9B7B997CFF9CDC725DC7F8A3054D5603A863EDA9DD2A19C80BE1D975A1
24A6AAD30EDF68D7BCBEB7677D60AC30B77244B0E642027DF157A1E62191E73C0380013C4CC1637CF320657B3E8743
E4072A6AF923F550DEABDDDE32E7E3993B4363737E0C21199D3BE3E93C27774C88E07
00 8C 00 60 80
7EE0068F1F40563006B86799780B5B3E0ADF9243DA2A51FE854CF8032C5E1DE45EE290757E68D7F5FBFEDD53E7B357
C435DA908D3C5250F5842BD3178DE39C063D1B0E01B094537FA5A5E8AB8E0ABB8E8E996A517690DCE42F0E0C5245AD
052E880B6BCEAE6B6E9DB4C387280D2C7EAA826F1BA91CC5370D46E138E5C91860D5
// MODULO, PublicExponent
00 8C A5 00 FF
EFF4538809B418C9FB7D6EFC8E6528CFA3DAF500B992D7F8CED705933D3E230590DD54C2E2A5BCE4C1F0F48F716D95
B43A0E9AFB6D4F3E1CF4CB8F2C7D9D10E27104A89817419296E93D6FF1D6B9900CC8D7C82D5653C446762C6BFC4B47
0AF36D6B997C9D272D0F62AB698997F4433B5226C6E804B13202108ECAF8CB9253B35C96213C75B54E696DA2E8A086
10698D3D4A702DEE6DE5BA450DD82BD5120C33732858EF5F0CFF7CEFA571167EFFDE517EB8F48E1A477582A1F3DCE5
A6BDE2E5FFF7569BB8B29223ADE10BF6D0D1C4B1AECB42EDB5D9B4FE5C02F1CB7E2F3A346D11E29D6DB48D3CBFEF621
F83525296EE3B29D102FB9703EBF50F7E929AF59
00 8C 00 10 03 01 00 01
//
00 84 02 20 00
00 84 03 20 00
00 84 04 20 00
00 84 05 20 00
00 84 06 20 00
00 84 00 20 00
00 84 01 20 00
//
// Sign
00 80 00 20 20 1234567890ABCDEF 1234567890ABCDEF 1234567890ABCDEF 1234567890ABCDEF
//
// /Q

```

```

Loading script test_apdu_rsa.txt ...
// select
Sending
00A4040006010203040500
Received
9000
// Admun PIN
Sending
00200001083030303030303030
Received
9000
//STATUS
//332E3030 F70A 1030 08C0 9000
Sending
0087000000
Received
332E3033F70B103005909000
// CLEAR RSA 0
Sending
0081002000
Received
9000
// GEN RSA 0
Sending
0082002000
Received
9000
// get key
Sending
0084002000
Received
A15FA5CAB4E8567E9F079C2D0705F3AFC785C9EE5829092B0B407CC955A068D7
4928A84BDA6E1FCE2D957AE60F01CD3F5A2949719468FAFB01411393446C8167

```

FC42A92FF6AB4586C5142EEEE53E6A6E2305B219E862B6C0DEA323352F8D4C28
9B2C2AE5BBA4780ACA9A016E2EA78F55CC275F94EE1A920F0D3670FA70A2CB5F
92B8D4A132847159E51AB11F6913050A014ECA6C268E81288E108F8636A00F75
AA7D95CD63D5799FB052738DB794A054143DF5073C3FFBE9CD6AFF3269EE7089
86E5885928D589581A993C0577369221A1C1D4E40D1E0737B8E84A3425BA42FE
AD02026889E790025626F353CB1DCA95C4E18AF7DD9B3D7C51A268E4B0A72305
9000
Sending
0084012000
Received
0100019000
Sending
0084022000
Received
AFD98008A26C18007CF4890B37720EB801286D82BC96515131E79576DC787DC5
FE63C1C6FE3541E5BFEFBC3DFDE81070CF8675910E430C29F57BB1EBDDA3150E
D4882E88780C89448B6E1975991A225E91EBFF820AF95AC4955A58C30C17517D
DEAAA65909E03D277D9428113B76246398A8177ADFA4943AE5E4980D2EEA97CB
9000
Sending
0084032000
Received
EAE0DF252DE0143C91F014A8DEF630CA2C35E0D862275069B367D9CE78C4B1CC
F1822BE51381A3EBCBD6B967724EB021697A2D1C322E3295BAB2D8A73E7B3EBF
1FD3DB952798AEA2C6E3755DED53F31F2A5499286850D5BBC0AE34A1DBCA7423
565E49ED68F2C8DDD143E905E5E4EF890CC8242220A6260949C10578406FB66F
9000
Sending
0084042000
Received
4304C0B009BEEA92DD0C91624DFA5133BD753BEF549271616184A2490F937853
CA3518B80F501AD246094713D53DE96E2519589B34F546A083276652BF409518
F6D503EF0CD9B2430664AA9D50014ED1FE091472EA5DC8431DACB62177ED3728
A281252ECFFDBF30DAC26956CFF3A1C0B780709A6ABB9DA6A8D12B45F2F8836B
9000
Sending
0084052000
Received
421D6D41BD636ECE8D13CFFCA6A30AA01B491E7A474561301E397FE6C7BE337B
C60C286653D30C5AF406FE2ECAD0B268F4D25997B1D7A68CD4FA3856F9129E5A
138F00213DFF8E46694AE26108C5B38A75D351373A0465E26D1B99DC638ADA9C
875609E032C5011ACC261C92785CB1BD1447CF437B84C6F79E36205FF3627D29
9000
Sending
0084062000
Received
AE35F4CCE2816C4EBFFB7B6DF18323A20AF41F47BC27C468178B73F892FF9AE3
4BBA44BE089FF3FA1B85A00380DFB0594A210E9962E4CC38FCB81F55863573F7
DDBE53218745A10D996C33ED1F30F7D977952A132DC2AC61C40095C596A6941B
0B20CF7D7D5C1FE4700B283739F22C7589EA8EDFC69657CD8CD480115DB795C2
9000
//
// Sign
Sending
00800020201234567890ABCDEF1234567890ABCDEF1234567890ABCDEF123456
7890ABCDEF
Received
800A532A3512AC42A7E66A39459B7B295FDF9A6D29B4046226FE98322DEFC167
DDAB0339E6805C1ED6623FA2F3935D01D9D8E58FF4846DC6577428A9995F2413
739AB1074D393D47624D9FD7EE90D062C93182726B1A23EE487829B3FF9856D7
96BB6352487C3DF69D4447FC4797FCA38E3DA3C65DCE5B6286F88E4457A7ACDA
E6701C6BB21FE1E5572DE1CD430AEA0C709C98F421152C303D85DFA0EB40C5D7
19FE1BECD8302623399306B4B38A942ACC135F5ABC9ED9FBBBD37C676DCD777C6
7F90BAD4284D0C47CF951952440B408B28486366C050F112948ACABE7077BB34
43006F8726A475F8278B6731DED055498ADCD293B24FAD3A50A83C9889FD5C92
9000
//
// CLEAR RSA 0
Sending
0081002000
Received
9000
// GEN RSA 0
Sending
0082002000
Received
9000

```
// get key
Sending
0084002000
Received
A537157C2609E026AEBF374DDDB1C7734FB17214B3C2742F1384BB96F4F00536
C08903A85CDFA28F1A6F6D61898323BD1CC944C3A48E0D4AEE1E0866125C8C27
3C173134AD45D09E4C513E47CDD13ECDB18FA2C153D5EAF1BAF38815C7FCE6C
3941B5C82BA80A282A5B96A2614566446A5CBE90405D97560F6BA994FEC7AE73
CEA9927E7B429ACE3B9BCC83F7B1C9C7BF7920DF80E57C6361C86DF7C9668542
EC608A86C96148BF40ADC7557FE7E05767183F59D2404080B910F58170B309DD
58A3FFC39C960DE85573EDA7BFD4E41556BBE80B916EB65AFBFEC91FD4C1E7A0
A780E107CF524053156772EDBA777BB9CFA182D0BCDADE787CC204B95CB492AD
9000
Sending
0084012000
Received
0100019000
Sending
0084022000
Received
A944F36A43AEA9E8FBBE003164E14C528A27984B41FA6AA6735397A0A0D8DC51
1883C13C16845350C69F62258D98CC0C9C3DA1482E63144C24BED36F1FE0BB4A
0A2AE166740C69A2CC33D65535012F40251108BCB0D0C7ED9FF5D293E0378BB1
D6DBC297C26AC29A37A6F8C20A29157777DF0DB1A459BF1A31460679C40842B
9000
Sending
0084032000
Received
F9DE5920DE0945B760373ADE2841350DBA247A40F1E6DEE3F2B6A00EC94330DB
19BA96DBBF3CDBB25BC545E3F1B00B00F3EC775577823501B03484EE95951253
7BF7489B40957E9DEFF48BBBA416CF8E8997B534E8DA51769331990DD84962C0
F5426C4050EBDFF783D3E640A7A9C1913427C97D5594409CD802EC84FE3CA087
9000
Sending
0084042000
Received
4A3D7BA577D9FB9BBC0E3177788F78321B76124DAFBA6D5C63E89DF04D9BCFC5
A1C2E7D5BAEA7EC046DD50E6B79BF4A2F343F4A0F768C79CB63575DFBD6967
A7DDF6CE40C7D0ED1E32FE269100DC007E41625BA011721EBEB1CA92211D3B9F
DC64BA4E8FAD46855CB62E867F29CE2556BB5E103B0D9CBFF68ED159A0268BD1
9000
Sending
0084052000
Received
BD56D49908476471C151F8A4B142402E87B7EE7FDA0E4525DC1CB10AA34DEBEE
8AFF8E233435AC90E9E381C369B8298E6E47E861762ECF75D90CE3125B7CDBA7
314FBFD00D4F7AE3B919C8ECC63996CBFECD5BBD9D191A13CD140F53A36A708F
A109D537A35A299BC864409D0BAFA20322CF676A36F82141F899B69A4C80AA6F
9000
Sending
0084062000
Received
394295313F2DD5A851FEF63165B7566677964998B60FDD378AA3FBB74F575262
86D78A331807071F0EE0AE8DB293CD9DA05A2A8C4D3046EE67183551F59B5BCD
CB4BC23B0DFA9361AD5891BE143F869149136B4E1A4192B8AC3AC490848445A2
DB6CC3C8A45D67151FA9A0064A4652455EA6F85C66147EC2E1A13485F01F38E2
9000
//
Sending
0081002000
Received
9000
// P, Q, DP1, DQ1, PQ
Sending
008C002080A6DBBA1977E559876A3F42A1AFFD57AA1F402494E22BA17523E4F2
729D3145DFBCDE29431A774C30DE6D09CD8DF8299F1355CFEEC9A3824711C1F3
E4B3FFCE02668DEF4CA5085873E10C59C2E68074C208A04297CE081F56679E09
231970D303A56238F7ABF0CB2E09B8BF9A51346EEE3D90A3E2346B83A1D3ED00
FFCD36AF6F
Received
9000
Sending
008C003080FE9644F0A6D0EF464238F38B53AF907474C5B12FE637E0095851F1
7AB1FEA9BD9C745AE43BAEB0664969C2E5B09121DC07D72235679CE7B36BB50A
F74B2258724CE5D4E15F7EAE9D7A21F398EF1634606504B842A6076D6E0CD9B0
4A3E0116063EBC3F949324849B2409CD428D2D68959CA4C231D76FC16BB6AAA7
EB35DEA9B7
```

Received
9000
Sending
008C0040808288C6F001B1F4F651122848616BAA5736977186DFE10E309BE212
21C403B60C30AF044CB0B9BF6B43B6BDC2875C2B037D96D65292804DFCD7518A
41D2E73DFCF6C9F6F958EA4C786B08C4780AE757702C5B637D6AFB43B8CEDAFD
7EEDFA2BBA607994E37D29A569E0C51D6FD63354CD2070B524D3A98092EB3E30
57EDECE815
Received
9000
Sending
008C0050801AD2C856B6ACC2D91B3716E9F21A48550CCC9B7B997CFF9CDC725D
C7F8A3054D5603A863EDA9DD2A19C80BE1D975A124A6AAD30EDF68D7BCBEB767
7D60AC30B77244B0E642027DF157A1E62191E73C0380013C4CC1637CF320657B
3E8743E4072A6AF923F550DEABDDE32E7E3993B4363737E0C21199D3BE3E93C2
7774C88E07
Received
9000
Sending
008C0060807EE0068F1F40563006B86799780B5B3E0ADF9243DA2A51FE854CF8
032C5E1DE45EE290757E68D7F5FBFEDD53E7B357C435DA908D3C5250F5842BD3
178DE39C063D1B0E01B094537FA5A5E8AB8E0ABB8E8E996A517690DCE42F0E0C
5245AD052E880B6BCEAE6B6E9DB4C387280D2C7EAA826F1BA91CC5370D46E138
E5C91860D5
Received
9000
// MODULO, PublicExponent
Sending
008CA500FFEFF4538809B418C9FB7D6EFC8E6528CFA3DAF500B992D7F8CED705
933D3E230590DD54C2E2A5BCE4C1F0F48F716D95B43A0E9AFB6D4F3E1CF4CB8F
2C7D9D10E27104A89817419296E93D6FF1D6B9900CC8D7C82D5653C446762C6B
FC4B470AF36D6B997C9D272D0F62AB698997F4433B5226C6E804B13202108ECA
F8CB9253B35C96213C75B54E696DA2E8A08610698D3D4A702DEE6DE5BA450DD8
2BD5120C33732858EF5F0CFF7CEFA571167EFFDE517EB8F48E1A477582A1F3DC
E5A6BDE2E5FFF7569BB8B29223ADE10BF6D0D1C4B1AECB42EDB5D9B4FE5C02F1
CB7E2F3A346D11E29D6DB48D3CBEF621F83525296EE3B29D102FB9703EBF50F7
E929AF59
Received
9000
Sending
008C001003010001
Received
9000
//
Sending
0084022000
Received
A6DBBA1977E559876A3F42A1AFFD57AA1F402494E22BA17523E4F2729D3145DF
BCDE29431A774C30DE6D09CD8DF8299F1355CFEEC9A3824711C1F3E4B3FFCE02
668DEF4CA5085873E10C59C2E68074C208A04297CE081F56679E09231970D303
A56238F7ABF0CB2E09B8BF9A51346EEE3D90A3E2346B83A1D3ED00FFCD36AF6F
9000
Sending
0084032000
Received
FE9644F0A6D0EF464238F38B53AF907474C5B12FE637E0095851F17AB1FEA9BD
9C745AE43BAEB0664969C2E5B09121DC07D72235679CE7B36BB50AF74B225872
4CE5D4E15F7EAE9D7A21F398EF1634606504B842A6076D6E0CD9B04A3E011606
3EBC3F949324849B2409CD428D2D68959CA4C231D76FC16BB6AAA7EB35DEA9B7
9000
Sending
0084042000
Received
8288C6F001B1F4F651122848616BAA5736977186DFE10E309BE21221C403B60C
30AF044CB0B9BF6B43B6BDC2875C2B037D96D65292804DFCD7518A41D2E73DFC
F6C9F6F958EA4C786B08C4780AE757702C5B637D6AFB43B8CEDAFD7EEDFA2BBA
607994E37D29A569E0C51D6FD63354CD2070B524D3A98092EB3E3057EDECE815
9000
Sending
0084052000
Received
1AD2C856B6ACC2D91B3716E9F21A48550CCC9B7B997CFF9CDC725DC7F8A3054D
5603A863EDA9DD2A19C80BE1D975A124A6AAD30EDF68D7BCBEB7677D60AC30B7
7244B0E642027DF157A1E62191E73C0380013C4CC1637CF320657B3E8743E407
2A6AF923F550DEABDDE32E7E3993B4363737E0C21199D3BE3E93C27774C88E07
9000
Sending

```

0084062000
Received
7EE0068F1F40563006B86799780B5B3E0ADF9243DA2A51FE854CF8032C5E1DE4
5EE290757E68D7F5FBFEDD53E7B357C435DA908D3C5250F5842BD3178DE39C06
3D1B0E01B094537FA5A5E8AB8E0ABB8E8E996A517690DCE42F0E0C5245AD052E
880B6BCEAE6B6E9DB4C387280D2C7EAA826F1BA91CC5370D46E138E5C91860D5
9000
Sending
0084002000
Received
A5EFF4538809B418C9FB7D6EFC8E6528CFA3DAF500B992D7F8CED705933D3E23
0590DD54C2E2A5BCE4C1F0F48F716D95B43A0E9AFB6D4F3E1CF4CB8F2C7D9D10
E27104A89817419296E93D6FF1D6B9900CC8D7C82D5653C446762C6BFC4B470A
F36D6B997C9D272D0F62AB698997F4433B5226C6E804B13202108ECAF8CB9253
B35C96213C75B54E696DA2E8A08610698D3D4A702DEE6DE5BA450DD82BD5120C
33732858EF5F0CFF7CEFA571167EFFDE517EB8F48E1A477582A1F3DCE5A6BDE2
E5FFF7569BB8B29223ADE10BF6D0D1C4B1AECB42EDB5D9B4FE5C02F1CB7E2F3A
346D11E29D6DB48D3CBEF621F83525296EE3B29D102FB9703EBF50F7E929AF59
9000
Sending
0084012000
Received
0100019000
//
// Sign
Sending
00800020201234567890ABCDEF1234567890ABCDEF1234567890ABCDEF123456
7890ABCDEF
Received
403B383E773512176786574CE741AD51133E9AC5866E7CFE7A3DBE47453E0095
DC0C505DC004FA8445B2AA96DD1E04ED1DED4CC681B13BB1D7A2A8A246EA1B63
598ECFBF593A3F508AA55F37E4B1F91DEECE3EAC1019E1D542CDFE03247C3F04
90A8D39E1EF7A3D443B6A9B3535A2B7B7DA9978109BFECB4CD05F45ED52C0001
B961D2C706B7F3710863DED1F1241399C08C09209B760AA596605BD9446040BF
3D8C7E8187C659A47415724BC6311E859CDFACD7A0E6FDA9D453172975101270
EBAB183E251510DD8A0A5D54BE9B4414F171BD1AF111292A38C06ED841B7F5F6
182FC54A512CFD15810E6BAF17525865F3D0D8771522235F887C7BD7503C5CB8
9000
//
Time: 9122 ms

```

3.6 test_apdu_connect

```

// Select
00A4040006010203040500
// ADM PIN
00200001083030303030303030
// TEST_DH
00DA0001206400FFBEDDCC892726FE239EA1A114BA4F35E713197FDA3708518EECA42679F8
// TEST_PK
00DA000241
040B5A6EC31366D6FC7ACB5C0E6EF15335D4DE88B04A5E4CE493E6B8FFB46B243F3261FAB90AA141A5FE646410204F
EA2DC45823710AE71606A38D2E42A7D574FF
// TEST_RND
00DA0003203e098dc904020111cfe180cdf3a9d699e7e3b22599f4f9793a06ba124827c475
//
// ClientTx: iv: 92A398153F423ADE29339CB8 key: D6EDC4AA3F50F38B2816A99DD162D935
// ClientRx: iv: 48E08797688ED8C99555434F key: 52A45CC9319E594B58D47BE5208CF86D
//
// AESCCM Decrypt
0085050639 D6EDC4AA3F50F38B2816A99DD162D935 92A398153F423ADE29339CB8 1703030018
03e21dc525c8c10603a54badb163a0c37933ae36158be33a
// 1703030018 47455420F0D0A17 9000 "GET /CrLf"
// AESCCM Encrypt
0085050648 52A45CC9319E594B58D47BE5208CF86D 48E08797688ED8C99555434F 1703030027
a69ec5fb37e335aaa37f4ef45e67d27daac7d8117e801ffee7b74d0b232970b5c3593ddb61bdc
// 1703030027 49206865617220796F75206661207368697A7A6C6521 17 9000 "I hear you fa shizzle!"
// AESCCM Encrypt
0085050527 D6EDC4AA3F50F38B2816A99DD162D935 92A398153F423ADE29339CB8 1703030016 3F30300D0A 17
//1703030016 7B9779E800D280903191436413CE3C0D665A5F1DC73A 9000
// AESCCM Decrypt
008505064C 52A45CC9319E594B58D47BE5208CF86D 48E08797688ED8C99555434F 170303002B
AACAC5FB24E567A6BF7E4EF95A3ED261ACCF75D288F384C49D040BD2B46EFB9B288CEB2306E447B2E6798

```

```

//170303002B 45746865727472757374206B657973746F726520332E3033 0D0A 17 "Ethertrust keystore
3.03CrLf"
// Clear Key3 P2=KeyIndex
0081000300
// init curve #1 = SECP256v1 key3
0089000300
// KeyPriv Client #3
00880703 20 E26374EBEDD02F6473918C2B645055E1AF3F174D7746301AEA63690E6900D554
// KeyPub Client #3
00880603 41
040B5A6EC31366D6FC7ACB5C0E6EF15335D4DE88B04A5E4CE493E6B8FFB46B243F3261FAB90AA141A5FE646410204F
EA2DC45823710AE71606A38D2E42A7D574FF
// TEST DHE(KeyPub Server)
008A0003 41
049205b4f3d7d7293c965d253edec44adc4e9af719d23502c58d661a04e562769613a15fe3ccbc2364c268bdd02d11
ee0cf49c1affbc2feb902e9724b650eefe8c
// set PSK2...
// 0085FF0A230100200102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
// reset tls
00D8000000
// ClientHello
00D800012016030101210100011D0303505B562E5ADC969EE305C92B85D2CBE5380938079A
00D800002099F16FFA7E09D9607E47682025C41720E154AF5D6A7AA4742701C898C1FD89D2
00D80000202206D4D3954C01D87B45290F0004130400FF010000D0000B000403000102000A
00D80000200004000200170016000000170000000D001E001C040305030603080708080809
00D8000020080A080B080408050806040105010601002B0003020304002D00020101003300
00D800002047004500170041049205B4F3D7D7293C965D253EDEC44ADC4E9AF719D23502C5
00D80000208D661A04E562769613A15FE3CCBC2364C268BDD02D11EE0CF49C1AFFBC2FEB90
00D80000102E9724B650EEFE8C0029003A0015000F
00D8000020436C69656E745F6964656E746974790000000000212016692C350E8CD52EE726
00D801021602357509DFCFDE1F6B69106C34A9B2BE575935B732E
// 00C0000010
00C000001C
00C000003A
// CCS (optional)
00D8000306140303000101
// Client Finished
// Tx: "00CrLf"
00D800033A1703030035cfddc75fc39b3a537b257744856c22df99d1e3e09065828e56759f9fa41e6b8adec085e630
2a6fb8d4beaead7503574690745f9f07
// Rx: "Ethertrust keystore 3.03CrLf"
00D800031B 1703030016 7B9779E800D280903191436413CE3C0D665A5F1DC73A
//170303002B
AACAC5FB24E567A6BF7E4EF95A3ED261ACCF75D288F384C49D040BD2B46EFB9B288CEB2306E447B2E6798 9000

```

```

Loading script script.txt ...
// Select
Sending
00A4040006010203040500
Received
9000
// ADM PIN
Sending
00200001083030303030303030
Received
9000
// TEST_DH
Sending
00DA0001206400FFBEDDCC892726FE239EA1A114BA4F35E713197FDA3708518E
ECA42679F8
Received
9000
// TEST_PK
Sending
00DA000241040B5A6EC31366D6FC7ACB5C0E6EF15335D4DE88B04A5E4CE493E6
B8FFB46B243F3261FAB90AA141A5FE646410204FEA2DC45823710AE71606A38D
2E42A7D574FF
Received
9000
// TEST_RND
Sending
00DA0003203E098DC904020111CFE180CDF3A9D699E7E3B22599F4F9793A06BA
124827C475
Received
9000
//

```

```

// ClientTx: iv: 92A398153F423ADE29339CB8 key: D6EDC4AA3F50F38B2816A99DD162D935
// ClientRx: iv: 48E08797688ED8C99555434F key: 52A45CC9319E594B58D47BE5208CF86D
//
// AESCCM Decrypt
Sending
0085050639D6EDC4AA3F50F38B2816A99DD162D93592A398153F423ADE29339C
B8170303001803E21DC525C8C10603A54BADB163A0C37933AE36158BE33A
Received
1703030018474554202F0D0A179000
// 1703030018 474554202F0D0A17 9000 "GET /CrLf"
// AESCCM Encrypt
Sending
008505064852A45CC9319E594B58D47BE5208CF86D48E08797688ED8C9955543
4F1703030027A69EC5FB37E335AAA37F4EF45E67D27DAAC7D8117E801FFEEFB7
4D0B232970B5C3593DDBD61BDC
Received
170303002749206865617220796F75206661207368697A7A6C6521179000
// 1703030027 49206865617220796F75206661207368697A7A6C6521 17 9000 "I hear you fa shizzle!"
// AESCCM Encrypt
Sending
0085050527D6EDC4AA3F50F38B2816A99DD162D93592A398153F423ADE29339C
B817030300163F30300D0A17
Received
17030300167B9779E800D280903191436413CE3C0D665A5F1DC73A9000
//1703030016 7B9779E800D280903191436413CE3C0D665A5F1DC73A 9000
// AESCCM Decrypt
Sending
008505064C52A45CC9319E594B58D47BE5208CF86D48E08797688ED8C9955543
4F170303002BAACAC5FB24E567A6BF7E4EF95A3ED261ACFC75D288F384C49D0
40BD2B46EFB9B288CEB2306E447B2E6798
Received
170303002B45746865727472757374206B657973746F726520332E30330D0A17
9000
//170303002B 45746865727472757374206B657973746F726520332E3033 0D0A 17 "Ethertrust keystore
3.03CrLf"
// Clear Key3 P2=KeyIndex
Sending
0081000300
Received
9000
// init curve #1 = SECP256v1 key3
Sending
0089000300
Received
9000
// KeyPriv Client #3
Sending
0088070320E26374EBEDD02F6473918C2B645055E1AF3F174D7746301AEA6369
0E6900D554
Received
9000
// KeyPub Client #3
Sending
0088060341040B5A6EC31366D6FC7ACB5C0E6EF15335D4DE88B04A5E4CE493E6
B8FFB46B243F3261FAB90AA141A5FE646410204FEA2DC45823710AE71606A38D
2E42A7D574FF
Received
9000
// TEST DHE(KeyPub Server)
Sending
008A000341049205B4F3D7D7293C965D253EDEC44ADC4E9AF719D23502C58D66
1A04E562769613A15FE3CCBC2364C268BDD02D11EE0CF49C1AFFBC2FEB902E97
24B650EEFE8C
Received
6400FFBEDDCC892726FE239EA1A114BA4F35E713197FDA3708518EECA42679F8
9000
// set PSK2...
// 0085FF0A230100200102030405060708090A0B0C0D0E0F101112131415161718191A1B1C1D1E1F20
// reset tls
Sending
00D8000000
Received
9000
// ClientHello
Sending
00D800012016030101210100011D0303505B562E5ADC969EE305C92B85D2CBE5
380938079A

```

```

Received
9000
Sending
00D800002099F16FFA7E09D9607E47682025C41720E154AF5D6A7AA4742701C8
98C1FD89D2
Received
9000
Sending
00D80000202206D4D3954C01D87B45290F0004130400FF010000D0000B000403
000102000A
Received
9000
Sending
00D80000200004000200170016000000170000000D001E001C04030503060308
0708080809
Received
9000
Sending
00D8000020080A080B080408050806040105010601002B0003020304002D0002
0101003300
Received
9000
Sending
00D800002047004500170041049205B4F3D7D7293C965D253EDEC44ADC4E9AF7
19D23502C5
Received
9000
Sending
00D80000208D661A04E562769613A15FE3CCBC2364C268BDD02D11EE0CF49C1A
FFBC2FEB90
Received
9000
Sending
00D80000102E9724B650EEFE8C0029003A0015000F
Received
9000
Sending
00D8000020436C69656E745F6964656E746974790000000000212016692C350E
8CD52EE726
Received
9000
Sending
00D801021602357509DFCFE1F6B69106C34A9B2BE575935B732E
Received
16030300A10200009D03033E098DC904020111CFE180CDF3A9D699E7E3B22599
F4F9793A06BA124827C4752025C41720E154AF5D6A7AA4742701C898C1FD89D2
2206D4D3954C01D87B45290F1304000055002900020000003300450017004104
0B5A6EC31366D6FC7ACB5C0E6EF15335D4DE88B04A5E4CE493E6B8FFB46B243F
3261FAB90AA141A5FE646410204FEA2DC45823710AE71606A38D2E42A7D574FF
002B000203049F1C
// 00C0000010
Sending
00C000001C
Received
1703030017715A3CD1CF78078F0F27C26DE797DD9FB55C7983B0E4B69F3A
Sending
00C000003A
Received
170303003570068C389205B036BD837F37AD2319DD3FFC0C3B59DFFFA74F7BBA
93B07154526086A490329DE746B02142D45713BC514A075242C39000
// CCS (optional)
Sending
00D8000306140303000101
Received
9000
// Client Finished
// Tx: "?00CrLf"
Sending
00D800033A1703030035CFDDC75FC39B3A537B257744856C22DF99D1E3E09065
828E56759F9FA41E6B8ADEC085E6302A6FB8D4BEAEAD7503574690745F9F07
Received
9001
// Rx: "Ethertrust keystore 3.03CrLf"
Sending
00D800031B17030300167B9779E800D280903191436413CE3C0D665A5F1DC73A
Received
170303002BAACAC5FB24E567A6BF7E4EF95A3ED261ACCF75D288F384C49D040

```

BD2B46EFB9B288CEB2306E447B2E67989000
//170303002B
AACAC5FB24E567A6BF7E4EF95A3ED261ACCF75D288F384C49D040BD2B46EFB9B288CEB2306E447B2E6798 9000
Time: 1948 ms

4 Test RACS

4.1 Test_racs_openssl

```
@echo off
openssl s_client -tls1_2 -CAfile root.pem -connect 127.0.0.1:7777 -ciphersuites
TLS_AES_128_GCM_SHA256 -no_ticket -cert client.pem -key clientkey.pem -pass pass:pascal
PAUSE

CONNECTED(00000190)
Can't use SSL_get_servername
depth=1 C = FR, ST = France, L = Paris, O = EtherTrust, OU = Test, CN = Admin, emailAddress =
pascal.urien@ethertrust.com
verify return:1
depth=0 C = FR, ST = IleDeFrance, L = Paris, O = ethertrust.com, CN = Server
verify return:1
---
Certificate chain
 0 s:C = FR, ST = IleDeFrance, L = Paris, O = ethertrust.com, CN = Server
  i:C = FR, ST = France, L = Paris, O = EtherTrust, OU = Test, CN = Admin, emailAddress =
pascal.urien@ethertrust.com
 1 s:C = FR, ST = France, L = Paris, O = EtherTrust, OU = Test, CN = Admin, emailAddress =
pascal.urien@ethertrust.com
  i:C = FR, ST = France, L = Paris, O = EtherTrust, OU = Test, CN = Admin, emailAddress =
pascal.urien@ethertrust.com
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIDfjCCAmagAwIBAgIJANOfJPKFO0bXMA0GCSqGSIb3DQEBCwUAMIGOMQswCQYD
VQQGEwJGUjEPMAGAlUECAwGRNjhmNlMQ4wDAYDVQQHDAVYXJpczETMBEGA1UE
CgwKRXRRO2ZJUcnVzdDENMAsGA1UECwwEVGVzdDEOMAwGA1UEAwWFQWRtaW4xKjAo
BgkqhkiG9w0BQCEW3Bhc2NhbC5lcml1bWVzZGhlcnRydXN0LmNvbTAwNDEy
MjMwODM1MzdaGA8yMTA3MDIxMjA4MzUzNlowXTELMAGAlUEBhMCR1IxFDASBgNV
BAgMC0lsZUR1RnJhbmNlMQ4wDAYDVQQHDAVYXJpczEXMBUGAlUECgwOZXRoZXJ0
cnVzdC5jb20xZDZANBGNVBAWMMB1NlcnZlcjCCAS1wDQYJKoZIhvcNAQEBBQADggEP
ADCCAQoCggEBALXQs0HXWNDEgiORikWmVK8q5VsIfphSLVyA3AJqcWoeiBnZDZMS
fpTYVAqcFov+R0fDDEvWib5lHHCKPWeJyZQ+K92p3ov3+8rukpmOah+M+yR3d/6B
8KmuDzUCMq9dfy+PvPLTrrloAib14QU/O8jsq2aSo9XtqTDuOCjIacRTqzi7VKWM
Al57qBIhHcfpM0debRNZUYEZgqp4br7aWkjNExWmJ7Chip4DlOYRyJjopYPODvPN
Qfn7KPKzqjTLTYGJhtfWoL2smej+JL+bsH3LBidiilv1cOqXuKJRI4r2o031DSPP
PkIVvC6cr9G6jTsqLYBOUuTfczEwGT2HFikCAwEAAaNMMAwCQYDVROTBAlwADAN
BgkqhkiG9w0BAQsFAAOCAQEAsNs131za/jwXKvc43HgUxhqr9XAqfbNO/Vzx/MDn
t54id5V7Y6+0bkZQ23q8i4znMZ3Z9/97+5A3dcOR5oaXQv8OnMYSNUXHeoqNX7rA
GD1G82eFaUtTXsfjPH/SxyFksmAzW+iM7+Z15KDgieVplmkG7rgViYefDPmYLNmd
JXqlcLb+ivgdPQrov9ZzDt+ZW450uS4bmCmAsxXWOFtr3ohvpeyewmB/F0HTMJmz
OEFP0113Za2+rXvQQAHA3G+BQTYEkan+IC27PaqywuXpvKAPedOusoDNfgelzNnA
ukiNgKOHsgsmP3xkVITZcSndvIgKI6kXuPY06CtZpFdb0A==
-----END CERTIFICATE-----
subject=C = FR, ST = IleDeFrance, L = Paris, O = ethertrust.com, CN = Server

issuer=C = FR, ST = France, L = Paris, O = EtherTrust, OU = Test, CN = Admin, emailAddress =
pascal.urien@ethertrust.com

---
No client certificate CA names sent
Client Certificate Types: RSA sign, DSA sign, ECDSA sign
Requested Signature Algorithms:
RSA+SHA512:DSA+SHA512:ECDSA+SHA512:RSA+SHA384:DSA+SHA384:ECDSA+SHA384:RSA+SHA256:DSA+SHA256:EC
DSA+SHA256:RSA+SHA224:DSA+SHA224:ECDSA+SHA224:RSA+SHA1:DSA+SHA1:ECDSA+SHA1
Shared Requested Signature Algorithms:
RSA+SHA512:DSA+SHA512:ECDSA+SHA512:RSA+SHA384:DSA+SHA384:ECDSA+SHA384:RSA+SHA256:DSA+SHA256:EC
DSA+SHA256:RSA+SHA224:DSA+SHA224:ECDSA+SHA224:RSA+SHA1:DSA+SHA1:ECDSA+SHA1
---
SSL handshake has read 2331 bytes and written 2920 bytes
Verification: OK
---
New, TLSv1.2, Cipher is AES128-GCM-SHA256
Server public key is 2048 bit
Secure Renegotiation IS supported
No ALPN negotiated
SSL-Session:
    Protocol : TLSv1.2
```

```

Cipher      : AES128-GCM-SHA256
Session-ID: AF3DEF26230A0DA4A01A34BF308B232ED050F9E218C6C4DBB53AE9C962EA8279
Session-ID-ctx:
Master-Key:
665ECB23A3829B286C576B2FA10BF0C8EFF1AE5576C6176CB52C32F8670C46E259ADD64C0F1045AC39D31E016F6688
65
PSK identity: None
PSK identity hint: None
SRP username: None
Start Time: 1788183061
Timeout      : 7200 (sec)
Verify return code: 0 (ok)
Extended master secret: no
---
BEGIN
LIST
END
BEGIN
+004 001 0 999
END

```